

SOLUTIONS MANUAL

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Chapter 1

Center of Mass

1.1 Solutions to Exercises

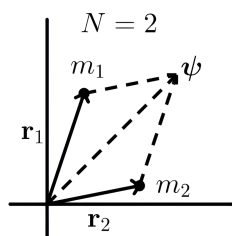


Figure 1.1

Q1.1

Consider a system S with $N = 2$ point masses (Fig. 1.1). Then, the total position-mass would be:

$$\boldsymbol{\psi} = m_1\mathbf{r}_1 + m_2\mathbf{r}_2$$

We can generalize the above result to a system with N points:

$$\boldsymbol{\psi} = m_1\mathbf{r}_1 + m_2\mathbf{r}_2 + \dots + m_N\mathbf{r}_N$$

Finally, we can more succinctly write the above summation as $\boldsymbol{\psi} = \sum_{k=1}^N m_k\mathbf{r}_k$

Q1.2

We'll let the object S be cut into slices, each with mass Δm at a position $\mathbf{r}$. Then, the total position mass would be:

$$\sum \Delta m \mathbf{r}$$

Next, we take the limit as the mass of each slice goes to zero so that $\Delta m \rightarrow dm$

$$\boldsymbol{\psi} = \lim_{\Delta m \rightarrow 0} \sum \Delta m \mathbf{r}$$

In the limit, the resulting summation becomes the integral $\boldsymbol{\psi} = \int_S \mathbf{r} dm$

Q1.3

The associative property of addition says that the grouping terms in parenthesis doesn't affect the final sum:

$$(a + b) + c = a + (b + c)$$

We can group the position-mass of multiple points into subsystems.

$$\boldsymbol{\psi} = (m_1\mathbf{r}_1 + \dots + m_i\mathbf{r}_i) + (m_{i+1}\mathbf{r}_{i+1} + \dots + m_n\mathbf{r}_n)$$

We let the first set be S_1 and the second S_2 so that $S_1 + S_2 \equiv S$. We then see

$$\boldsymbol{\psi} = \boldsymbol{\psi}_1 + \boldsymbol{\psi}_2$$

regardless of how we choose to split our original system S .

Q1.4

Let the discrete system S have a total mass:

$$M = \sum m_i$$

We defined the position-mass of a point P located at the center of mass to be equal to the system S given it has the same mass.

$$\psi_P = \psi_S$$

We then substitute the definition for the position-mass of a point and a system into each corresponding term.

$$M \mathbf{r}_{\text{CM}} = \sum \mathbf{r}_i \Delta m_i$$

We see that the center of mass is located at $\boxed{\mathbf{r}_{\text{CM}} = \frac{1}{M} \sum \mathbf{r}_i \Delta m_i}$

We can repeat the same process for a continuous object S with total mass:

$$M = \int_S dm$$

We equate the position-masses of the point P and the continuous system S , substituting their definitions:

$$M \mathbf{r}_{\text{CM}} = \int_S \mathbf{r}_i dm$$

We see that the center of mass is located at $\boxed{\mathbf{r}_{\text{CM}} = \frac{1}{M} \int_S \mathbf{r}_i dm}$

REMARK: We can visualize the center of mass as a special location in space where if we move all the mass there, the position-mass of the object doesn't change. It's the location where a point of equivalent mass equals the object.

Q1.5

We use the standard definition for the center of mass of a discrete system with masses m_1 and m_2 located at x_1 and x_2 (Fig. 1.2) :

$$\mathbf{r}_{\text{CM}} = \frac{1}{M} \sum_{i=1}^N \mathbf{r}_i m_i$$

$$N = 2$$

Then, we restrict the equation to two points (or $N = 2$) such that the total mass is given by:

$$M = m_1 + m_2$$

We see that the center of mass is located at $\boxed{\mathbf{r}_{\text{CM}} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}}$

REMARK: We were able to ignore the vector nature of the masses' positions because two points always lie on a line, meaning we can use scalars to describe their positions.

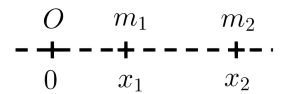


Figure 1.2

Q1.6

By the definition of superposition, we can split an object S into two parts, S_1 and S_2 such that their total position-masses' are related as follows:

$$\psi = \psi_1 + \psi_2$$

Let S , S_1 , and S_2 have masses m , m_1 and m_2 , which we can relate as:

$$m = m_1 + m_2$$

By the definition of the center of mass, we know the total position mass of each object equals the object's mass located at the center of mass. Therefore, let S , S_1 , and S_2 have center of mass vectors $\mathbf{r}_{\text{CM}}$, $\mathbf{r}_1$ and $\mathbf{r}_2$. (Fig. 1.3)

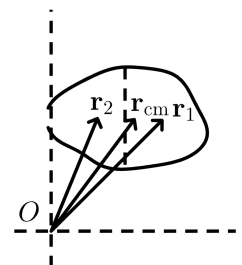


Figure 1.3

$$m\mathbf{r}_{\text{CM}} = m_1\mathbf{r}_1 + m_2\mathbf{r}_2$$

We isolate the center of mass of S and find $\mathbf{r}_{\text{CM}} = \frac{m_1\mathbf{r}_1 + m_2\mathbf{r}_2}{m_1 + m_2}$.

REMARK: This demonstrates that one can find the center of mass of an object by splitting the object into parts arbitrarily, then taking the weighted average of the parts. Make sure you understand how to generalize this to N parts. Also note that the vector nature of each part's center of mass cannot be ignored unlike the previous problem. Although the center of masses themselves may lie on a line, we made no assumption that both our coordinate axis is parallel to the line and our origin lies on the line.

Q1.7

Let the mass of a slice be m_i before being scaled and m'_i after being scaled by a factor k .

$$m'_i = km_i$$

We then find the new total mass:

$$M' = \sum m'_i = \sum km_i = k \sum m_i = kM$$

We then find the new center of mass:

$$r'_{\text{cm}} = \frac{1}{M'} \sum m'_i r_i = \frac{1}{kM} \sum km_i r_i = \frac{1}{M} \sum m_i r_i = r_{\text{cm}}$$

We can factor out the constant k , meaning the center of mass does not change.

REMARK: Consider a rod with the center of mass clearly lying halfway in between. It doesn't matter how dense or thick the rod is, the center of mass will always lie halfway in between. A similar principle applies to the general object S . Scaling each point also scales the total mass, having no effect on the center of mass. In the end, what matters is the rate of growth (or the differences in mass) between adjacent positions.

E1.8

We begin by finding the total position-mass of the system using the discrete definition. Recall that we must add each point's position-mass vector tail-to-tip.

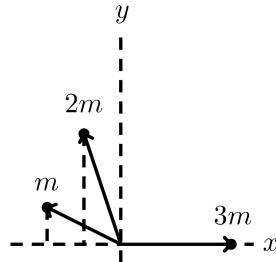


Figure 1.4

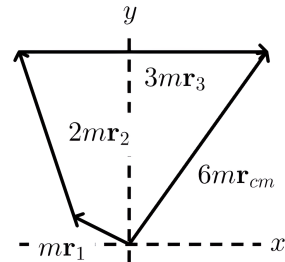


Figure 1.5

We will do this most easily decomposing our vectors into the x and y directions.

$$\psi_x = \sum m_i r_{ix} = (m)(-2d) + (2m)(-d) + (3m)(3d) = 5md$$

$$\psi_y = \sum m_i r_{iy} = (m)(d) + (2m)(3d) + (3m)(0) = 6md$$

Next, we find the total mass of the system:

$$M = \sum m_i = m + 2m + 3m = 6m$$

Finally, we set the position-mass of a single point with mass M equal to the total position-mass of the entire system.

$$Mr_x = \psi_x$$

$$Mr_y = \psi_y$$

Substituting and solving for the center of mass, we find $(r_x, r_y) = (\frac{5}{6}d, d)$.

REMARK: We see that the center of mass (Fig. 1.6) lies inside the triangle formed by the three points. It also lies slightly more towards $3m$ than $2m$ (although it's a bit hard to see) and $2m$ than m , meaning the distances from each point mass are ranked according to the masses of each point. Think about why these two properties should always hold true for a set of 3 points. Additionally, notice that in our calculation, $3m$ did not contribute to the vertical position-mass. However, it still contributed to the center of mass because it increased the total mass. Hence, our placement of the origin didn't affect the location of the center of mass.

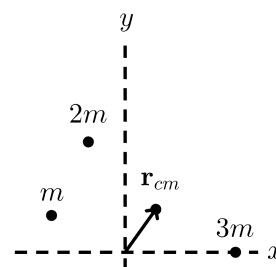


Figure 1.6

E1.9

Consider a point P_O with mass m located at (x, y) and two points P_x and P_y , both with mass m , located at $(x, 0)$ and $(0, y)$ respectively. We see that P contributes the same position-mass to the system as the combination of P_x and P_y .

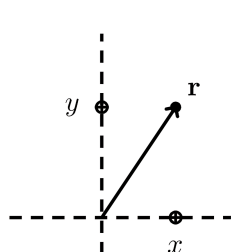


Figure 1.7

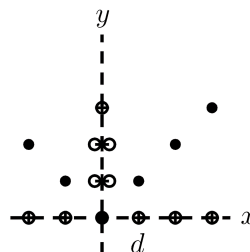


Figure 1.8

In essence, we can project any point onto the x and y axes respectively without changing the position-mass of a system. Knowing this, we can simplify our problem to a line of points located on the x and y axes.

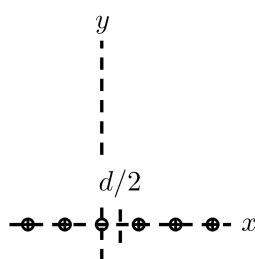


Figure 1.9

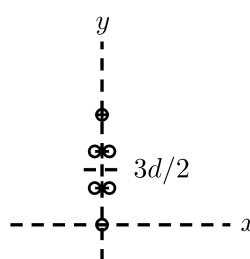


Figure 1.10

This quickly allows us to find the center of mass using symmetry as all the points have the same mass. We see the center of mass lies at $(r_x, r_y) = (\frac{1}{2}d, \frac{3}{2}d)$

REMARK: For sake of clarity, we put duplicate points slightly apart while they should lie directly on the y -axis. Although we could've solved this problem using the discrete definition, the conceptual idea of projecting mass onto the coordinate axes will be useful as we progress. Additionally, note that the center of mass doesn't lie on any of the points. Although many objects have a center of mass that lies within them, there is no reason why it has to.

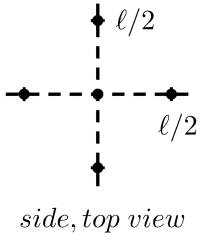


Figure 1.11

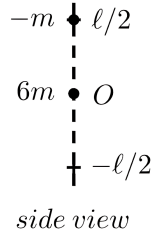


Figure 1.12

E1.10

We first add back the missing face, completing the cube. Using center of mass by parts, we can replace each face with a point mass m located at the center, allowing us to find the center of mass of the new cube, which simply lies at its center (Fig. 1.11) .

We can then set our origin to lie at the center of mass, making the total position-mass of the cube zero. Next, we can superpose a negative face with mass $-m$ where the missing face would have been. Doing so returns the system to its original state (Fig. 1.12) .

We see only the vertical position of the center of mass changes by symmetry. We can then solve for the new position-mass, that of our original system.

$$\psi_y = (-m)(l/2)$$

We then set the total position-mass equal to the position-mass of a point that has the same total mass as our system.

$$\psi_P = (6m - m)(r_y)$$

Equating our system and the point $\psi_y = \psi_P$, we find the center of mass to lie at $r_y = -\ell/10$.

REMARK: If we set our origin to be center of the bottommost face O' , we could shift our answer using vector addition:

$$r' = r + \Delta r$$

r' and r are the CM's relative to O' and O ; Δr is the vector from O' to O . This follows from vector addition.

$$r' = -\ell/10 + \ell/2$$

We see that the center of mass lies a distance $2\ell/5$ from the bottommost face.

E1.11

We add back the missing portion, completing the circle. The full circle has an area $A = \pi R^2$ while the added mini-circle has an area $A = \pi(R/4)^2 = \pi R^2/16$. We can assume the surface density σ of the circle to be constant, making their respective masses:

$$M = \pi\sigma R^2$$

$$m = \pi\sigma R^2/16$$

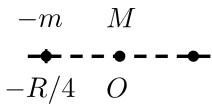


Figure 1.13

As before, we superpose a negative mini-circle where the hole previously was. We then use center of mass by parts with our origin located at the center of the larger circle (Fig. 1.13) .

$$\psi_x = (-m)(-R/4)$$

$$\psi_P = (M - m)(r_x)$$

We find the center of mass to lie at $r_x = \frac{1}{60}R$

REMARK: Notice that the ratio of M/m equals the ratio of the the radii of the larger to the smaller circle squared $(R/r)^2$. This arises from proportionality, which we'll explore in the next section.

E1.12

Let the i th point have a mass $m_i = m$ and a distance from the origin $r_i = id$. We can then apply the discrete definition for the position-mass of the system.

$$\psi = \sum_{i=1}^N m_i r_i = \sum_{i=1}^N (m)(id) = md \sum_{i=1}^N i = md \frac{N(N+1)}{2}$$

We then set the total position-mass of the line equal to that of a single point with equivalent total mass $M = mN$.

$$\psi = Mr_{cm} = mNr_{cm}$$

Solving for the center of mass, we find $r_{cm} = \frac{d(1+N)}{2}$.

REMARK: Notice that the center of mass lies at the average position of the first and last point due to symmetry. Additionally, we can find the center of mass of a continuous rod by fixing the distance of the furthest point to a constant $Nd = \ell$. Then, we can take the limit as the number of points goes to infinity $N \rightarrow \infty$ and the distance between each point goes to zero $d \ll \ell$.

$$\lim_{N \rightarrow \infty} r_{cm} = \lim_{N \rightarrow \infty} \frac{d(1+N)}{2} = \lim_{N \rightarrow \infty} \ell \frac{1/N + 1}{2} = \ell/2$$

As the line of points becomes a rod, the center of mass shifts backwards to exactly halfway in between the origin and the furthest point. That's because each point shifts back partially to fill in the empty space behind it.

Q1.13

A hollow cylinder C has axial symmetry (symmetry about its cylindrical axis) about an axis O , meaning that rotation about any angle θ results in an image C_θ congruent to the original. (Fig. 1.14)

$$C \equiv C_\theta$$

Let the C and C_θ have center of masses' $\mathbf{r}$ and $\mathbf{r}_\theta$. As the center of mass is an intrinsic property of an object, congruency of the object implies congruency of the center of mass.

$$\mathbf{r} \equiv \mathbf{r}_\theta$$

Let's assume the center of mass of C lies a distance d away and C_θ is formed after a rotation of 180° . As both lie on a line, we can substitute for their center of masses:

$$d = -d$$

We see the only solution becomes $d = 0$.

REMARK: We can work in the more general case of rotation about an angle θ and distance r . Let the preimage have a center of mass given as $\mathbf{r} = r\hat{i}$ allowing us to find the general form center of mass of the image (Fig. 1.15).

$$\mathbf{r}_\theta = r(\cos\theta\hat{i} + \sin\theta\hat{j})$$

Equating the components, we get a system of equations which we can solve:

$$r = r\cos\theta$$

$$0 = r\sin\theta$$

We see either $r = 0$ or $\theta = 0$. However, we know that the preimage and image are congruent for all rotations, meaning r must equal zero regardless of θ . Note that the above derivation extends the argument to shapes with rotational symmetry about only specific angles. This includes squares with 90° rotational symmetry and triangles with 120° rotational symmetry. We can plug the corresponding angle in the above system of equations, and note that $r = 0$ continues to remain the only solution. Hence, an object does not need to be invariant under all rotations, merely invariant under some rotations in order to constrain the center of mass.

Q1.14

Consider the general object S that has cylindrical symmetry but has density varying radially such that at any given radius, the density is constant. We can model this with a function as

$$\rho = f(r)$$

That means, we can slice S into cylindrical shells S_i , each with radius R_i and uniform density $\rho_i = f(R_i)$ (Fig. 1.16). By the previous exercise, we know that each

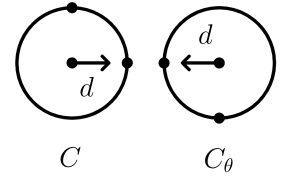


Figure 1.14

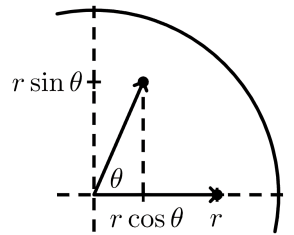


Figure 1.15

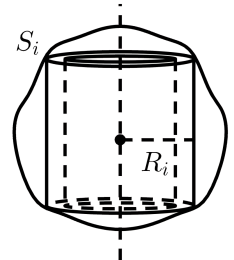


Figure 1.16

cylindrical shell has a center of mass r_i that lies on the cylindrical axis. This implies

$$r_i = 0$$

for all the shells because they are all concentric about the same axis. Then, we can find the center of mass by taking the weighted average of the parts. We'll solve for the radial component only as the vertical component varies for each slice.

$$\begin{aligned} M\mathbf{r}_{\text{CM}} &= m_1\mathbf{r}_1 + m_2\mathbf{r}_2 + \dots + m_n\mathbf{r}_n \\ &= m_1(0) + m_2(0) + \dots + m_n(0) = 0 \end{aligned}$$

We know that the $M > 0$ for any interesting object, so the center of mass indeed lies on the cylindrical axis. In other words, $\boxed{\mathbf{r}_{\text{CM}} = 0}$.

REMARK: In the above derivation, we only solved for the radial component of the center of mass, not the vertical component which varies with every slice. Additionally, the above result applies to any 3D object including cubes and pyramids with density varying radially. The only difference would be how we formed our shells, which would no longer be cylindrical.

Q1.15

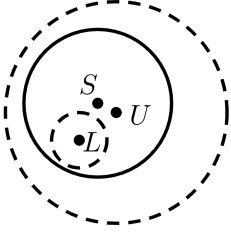


Figure 1.17

L and U stand for the lower and upper bound respectively, for the object S . However, the center of mass of L and U are points. We cannot bound a point between two points in space and in the plane (try it yourself). (Fig. 1.17) shows an example of why such a restriction would be ineffective. However, we can bound a point between two points given all three points lie on a line. That's because if center of S travels upwards, it reaches U , but it cannot cross U as U is the upper bound. If the center of S travels downwards, it reaches L , but it cannot cross L . Hence, we can declare the center of S lies between the centers of L and U . However, in the plane or space, the center of S can travel perpendicularly meaning there is effectively no restriction.

REMARK: The center of mass is not actually traveling. We are simply suggesting that using some intuition, we can create bounds for the center of mass to estimate whether our derived answer to a problem is reasonable. However, without cylindrical symmetry, those bounds would be practically ineffective (unless you only care about one component of the center of mass's position).

Q1.16

Let S have a center of mass $\mathbf{r}_{\text{cm}}$ and total mass M , meaning

$$M = \sum m_i$$

Let each slice S_i of S have a mass m_i and position $\mathbf{r}_i$ with respect to O . By the definition of the center of mass of S , we have

$$M\mathbf{r}_{\text{cm}} = \sum m_i\mathbf{r}_i$$

We then find the center of mass with respect to O' . Let the new position of each slice be $\mathbf{r}'_i$ and the new center of mass be $\mathbf{r}'_{\text{cm}}$

$$M\mathbf{r}'_{\text{cm}} = \sum m_i\mathbf{r}'_i$$

We then can rewrite each new position $\mathbf{r}'_i$ using its old position $\mathbf{r}_i$ and the vector from O' to O (Fig. 1.18) .

$$\mathbf{r}'_i = \Delta\mathbf{r} + \mathbf{r}_i$$

Finally, we substitute and utilize the identities established before.

$$\begin{aligned} M\mathbf{r}'_{\text{cm}} &= \sum m_i(\Delta\mathbf{r} + \mathbf{r}_i) \\ &= \Delta\mathbf{r} \sum m_i + \sum m_i\mathbf{r}_i \\ &= M\Delta\mathbf{r} + M\mathbf{r}_{\text{cm}} \end{aligned}$$

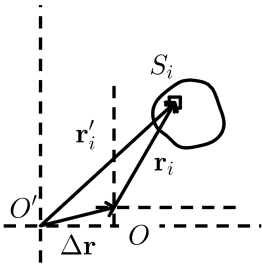


Figure 1.18

Where we utilized the fact that $\Delta \mathbf{r}$ is a constant. We see that the new center of mass lies at $\boxed{\mathbf{r}'_{cm} = \Delta \mathbf{r} + \mathbf{r}_{cm}}$

Q1.17

We first utilize the definition of the center of mass of S .

$$M\mathbf{r}_{cm} = \sum m_i \mathbf{r}_i$$

However, our origin O is located at the center of mass, meaning the center of mass vector is located from O to O . In other words,

$$\mathbf{r}_{cm} = 0$$

Finally, we see that the sum of all the position mass vectors evaluates to $\boxed{\sum m_i \mathbf{r}_i = 0}$

REMARK: We can see the center of mass is located at the origin in another way. Suppose the center of mass of S is $\mathbf{r}_c$ relative to O . Then, we change our origin to O' located at the center of mass. Hence, the vector from O' to O is $-\mathbf{r}_{cm}$. Then, we can utilize the previous exercises formula as $\mathbf{r}'_{cm} = \Delta \mathbf{r} + \mathbf{r}_{cm} = -\mathbf{r}_{cm} + \mathbf{r}_{cm} = 0$ to show the same result. This idea essentially postulates that the linear differences cancel around the center of mass.

Q1.18

Scaling an object by a factor k implies all the lengths are scaled by a factor k . Hence,

$$L' = kl \Rightarrow L \propto k$$

Recall the area of a square $A = s^2$. Any surface area can be approximated as built of many square slices. Hence,

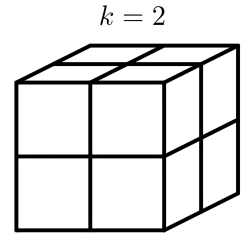
$$A' = (kl)^2 = k^2 l^2 \Rightarrow A \propto k^2$$

Finally, recall the volume of a cube $V = s^3$. any volume can be approximated as built of many cubical slices. Hence,

$$V' = (kl)^3 = k^3 l^3 \Rightarrow V \propto k^3$$

Let n represent the dimensionality of an object S . After S is scaled by a factor k , we see that the mass m_i of a slice S_i has a size s that scales by a factor $\boxed{s \propto k^n}$.

REMARK: (Fig. 1.19) intuitively shows the proportionality of lines, areas, and volumes to k , k^2 , and k^3 . The diagram shows $k = 2$; the length of each edge is twice as long, each face has consists of 4 smaller faces, and cube consists of 8 smaller cubes. Additionally, we claimed that any object can be approximated using many polygon slices (Fig. 1.20) regardless of the geometry of the object. This is only true in the limit where the number of slices goes to infinity ($N \rightarrow \infty$) as the unaccounted region (near the smooth edges) becomes negligible.

**Figure 1.19****Q1.19**

In the previous exercise, we showed that the slice S_i an n dimensional object S has a volume that scales by a factor n :

$$V'_i = V_i k^n \Rightarrow V_i \propto k^n$$

However, the density of the slice ρ_i also scales according to the density function $\rho(x)$, meaning

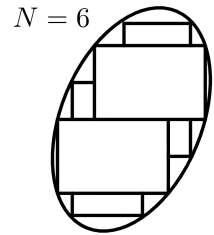
$$\rho'_i = \rho_i k^s \Rightarrow \rho_i \propto k^s$$

The mass of a slice is the density kg/m^3 multiplied by volume m^3 :

$$m_i = \rho_i V_i \Rightarrow m_i \propto \rho_i V_i$$

Combining the previous two results, we see that the mass of a slice scales by a factor

$$m'_i = V_i k^n * \rho_i k^s = m_i k^{n+s} \Rightarrow m_i \propto k^{n+s}$$

**Figure 1.20**

We can then add up the masses of all the slices in a object S noting $M = \Sigma m$.

$$M' = \sum m'_i = \sum m_i k^{n+s} = k^{n+s} \sum m_i = k^{n+s} M$$

We see that the mass of the object scales according to $M \propto k^{n+s}$.

REMARK: We can see that a rod ($n = 1$) with density varying quadratically according to $\rho = x^2$ ($s = 2$) has a mass that scales by the same factor as a cube ($n = 3$) with uniform density ($s = 0$). Think about why that is the case.

Q1.20

For convenience, we'll define r to be the sum of the dimensionality n and the exponent of the density function s .

$$r = n + s$$

Then, we sum up the position mass of all the slices S'_i of S_k given the initial position mass $\psi_0 = \Sigma m_i \mathbf{r}_i$.

$$\psi_k = \sum m'_i \mathbf{r}'_i = \sum m_i k^r * \mathbf{r}_i k = k^{r+1} \sum m_i \mathbf{r}_i$$

Substituting our initial position-mass, we see $\psi_k = \psi_0 k^{r+1}$.

REMARK: Notice the extra 1 which comes from the length of the position vector being scaled by a factor k too. Rather than memorize this formula, its more important to understand how we derived it. We considered three factors: Size, density, and the position vector of each slice. We analyzed the effects on each component to determine what happened to the total position-mass of the object.

Q1.21

We split S_1 into S_2 and S_3 , hence by superposition we have:

$$\psi_1 = \psi_2 + \psi_3$$

We know this is true for all S_2 and S_3 as long as they are parts of S_1 . We can then rearrange and take the limit without affecting the equality.

$$\lim_{S_2 \rightarrow S_1} \psi_3 = \lim_{S_2 \rightarrow S_1} \psi_1 - \psi_2$$

When S_2 becomes S_1 , S_3 becomes nothing. Hence, we have the limit $0 = 0$.

REMARK: Note that in the limit as S_2 becomes S_1 , the difference S_3 approaches the shell of S_1 . Position-mass is commonly written as $\beta m r$, where β represents the numeric coefficient tied to the geometry of the shape. The shell S_3 and the original shape S_1 have different geometries, hence they have different numeric coefficients. By clever algebraic manipulation, we can avoid $0 = 0$ and instead relate the two coefficients. If we know one, we can then solve for the other. This idea will become more clear as you solve some problems.

E1.22

We bound the hoop between a triangle and a rectangle, both missing their base. We can replace all the straight lines with points located at their centers with masses proportional to their lengths. The triangle's CM clearly lies at $R/2$.

Rather than solve for the rectangle, we can simply project all of its mass onto the y -axis. By symmetry, we can see the rectangle's CM lies at $3R/4$, halfway between $R/2$ and R (Fig. 1.21).

We can finally bound the CM r_{cm} of the hoop as $\frac{1}{2}R < r_{cm} < \frac{3}{4}R$.

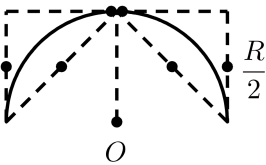


Figure 1.21

REMARK: The actual CM lies at $r_{cm} = \frac{2}{\pi}R \approx 0.637R$. The average of our bounds is $\bar{r} = \frac{1}{2}(R/2 + 3R/4) = 0.625R$. We see the CM lies between our bounds and is approximately halfway in between (Fig. 1.22). We expect this because any point on the triangle can be moved outwards to reach the circle, in the process moving upwards slightly. Similarly, any point on the circle can be moved outwards to reach the rectangle, in the process moving upwards slightly. The one-to-one correspondence with points gradually increasing in height makes the bounds work.

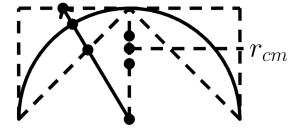


Figure 1.22

E1.23

We bound the circle between a solid, isosceles triangle and a rectangle. The center of mass of a rectangle clearly lies at $R/2$ while the problem tells us the center of mass of a isosceles triangle lies at $R/3$. We can then bound the center of mass r_c of a solid circle as $\boxed{\frac{1}{3}R < r_c < \frac{1}{2}R}$.

REMARK: The actual center of mass lies at $r_{cm} = \frac{4}{3\pi}R \approx 0.424R$ while the average of our bounds is $\bar{r} = \frac{1}{2}(R/3 + R/2) = 5R/12 \approx 0.417R$.

Q1.24

For a point ($N = 1$) located at the position $\mathbf{r}$, we get:

$$\mathbf{r}_{CM} = \frac{1}{1}(\mathbf{r})$$

For a line ($N = 2$) with vertices located at $\mathbf{r}_1$ and $\mathbf{r}_2$, we get:

$$\mathbf{r}_{CM} = \frac{1}{2}(\mathbf{r}_1 + \mathbf{r}_2)$$

We see that the center of masses are located at $\boxed{\mathbf{r} \text{ and } \frac{\mathbf{r}_1 + \mathbf{r}_2}{2}}$

REMARK: This makes intuitive sense. The center of mass of a point lies on the point itself and the center of mass of a line with uniform density lies halfway in between the two ends. However, if the density of a line was not uniform such that it was denser on one side, we would expect the center of mass to be lopsided, meaning halfway in between would no longer make sense. Note that we will assume uniform density unless a problem explicitly says otherwise.

E1.25

Because the density varies linearly, we can replace every point on the rod with a line such that the height of the line is proportional to the distance from the origin. Our line then becomes a triangle with constant density (Fig. 1.23). We now use the N -Points Theorem.

$$r_x = \frac{1}{3}(0 + \ell + \ell)$$

We see the center of mass of the rod lies at $\boxed{r_x = \frac{2}{3}\ell}$

REMARK: Recall that we can find the center of mass of an object by finding the center of mass of its parts and taking the weighted average of those parts. In essence, we're reversing the process here. We assume that each point on the rod is a center, and finding a part with uniform density that represents the center. Because the y -position is irrelevant to the rod, we ignore the height of each part.

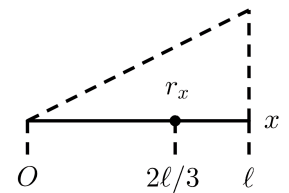


Figure 1.23

E1.26

In the previous exercise, density varied linearly allowing us to replace each point with a line of uniform density. Here, density varies quadratically. Hence, we can replace each point on the rod with a triangle with uniform density (Fig. 1.24). We now use the N -Points Theorem.

$$r_x = \frac{1}{4}(0 + \ell + \ell + \ell)$$

We see the center of mass of the rod lies at $\boxed{r_x = \frac{3}{4}\ell}$

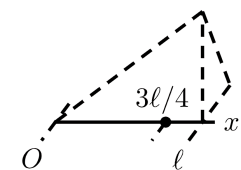


Figure 1.24

REMARK: Recall that the constant of each mass has no effect on the center of mass. Hence, using a triangle with area $A = \frac{1}{2}s^2$ to represent a point with density $\rho = x^2$ is valid as the rate of growth remains the same. We simply chose a triangle in order to apply the N -points theorem.

1.2 Solutions to Problems

1.1 N-Points Theorem Line

Recall the property that linear differences in position-mass cancel at the center of mass. That means we can move mass around as long as an equivalent amount of mass moves in the opposite direction and same magnitude relative to the center of mass.

moving mass to the center of mass doesn't change the total position mass of a system. Now suppose we have two points of mass m located a distance d apart.

$$m \quad m$$

Both masses can eject a particle of mass $m/2$ towards the center of mass of the two points (Fig. 1.25) .

$$\frac{m}{2} \quad m \quad \frac{m}{2}$$

The total position-mass doesn't change, meaning neither does the center of mass. Now, each particle can eject a mini-particle of mass $m/4$ towards the center point of each pair.

$$\frac{m}{4} \quad \frac{m}{2} \quad \frac{m}{2} \quad \frac{m}{2} \quad \frac{m}{4}$$

They can now all eject a particle of mass $m/8$ towards the center of each new pair. This process will continue indefinitely until we get a rod.

$$dm \quad dm \dots dm \quad dm$$

However, the total position mass never changes, and the slightly less mass on the edges becomes negligible as the number of splits goes to infinity. Hence, the center of mass of a rod lies at the average of the two endpoints.

REMARK: By simple symmetry, the center of a line would obviously lie at the halfway point, which would be the average of the endpoints. However, we'll use this technique for the next two shapes. For fun, you can try to think of way such that the endpoints have the same mass as the interior points on the line. What would have to be the ratio of the ejected masses?

1.2 N-Points Theorem Triangle

Solution 1: Suppose we begin with three points, each with mass m , such that the lines that connect them form a triangle. We can cut the triangle into triangular shells. All the shells have the same center because they are just scaled versions of each other. (Fig. 1.26)

That means, we only need to find the center of mass of one shell. We'll choose the outermost shell. Then, we'll take the limit as the number of shells goes to infinity, meaning the outermost shell essentially becomes 3 rods.

Now, we have 3 masses and 3 rods. Each mass can eject two particles with mass $m/4$, one in each direction, towards the center of mass of each pair. As the position-mass of each subsystem (each pair of points) doesn't change, the total position-mass of the entire system (the triangular shell) doesn't change either (Fig. 1.27) .

In the same method as the previous problem, all the points keep ejecting more points until the three points becomes a shell. The center of mass never changes. We see

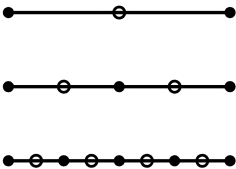


Figure 1.25

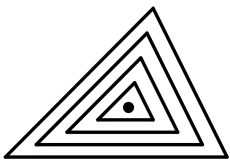


Figure 1.26

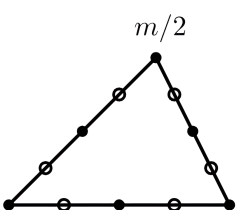


Figure 1.27

that the center of mass of a triangular shell is the same as that of three points. Hence, the center of mass of a triangle lies at the average of the three vertices.

REMARK: If you read through the proof carefully, you may feel we run into a contradiction. Supposing the averaging of the points method truly does work, then we have 3 lines with different lengths but the same mass. Although at first it may seem contradictory, it logically follows from how we cut our triangle. We cut our triangle such that in N slices, the triangle is complete. That means the combined width of N rectangles equals the distance from the centroid to each edge. However, the distance from the centroid to each edge is different for every edge. Hence, the width of each side of the slice must be different. From our reasoning, we see that the width of each rectangle that makes up a slice is inversely proportional to the length of the rectangle in order for all the sides of a slice to have the same mass. Try to think of another justification for the differing widths using geometry.

Solution 2: We can instead cut the triangle into rectangular strips such that each strip is parallel to one of the edges. We can take the limit as the number of strips goes to infinity so that each strip becomes a rectangular rod, with the center of mass lying exactly halfway in between (Fig. 1.28).

The resulting line formed by all the centers forms a median because it connects a vertex to the midpoint of the opposite side. We can then repeat this process with strips parallel to each of the three edges. The center of mass of a line of points obviously lies on the line. However, each median presents an equally valid method for finding the center of mass. That means the center of mass must lie on the point of concurrency of all three medians, known as the centroid.

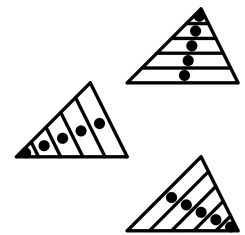


Figure 1.28

We now have to find a point that lies on all three medians. We claim that the average of the three vertices does. To do so, we imagine three vertices: A , B , and C , each with mass m . Let M be the midpoint of AB . We will find the center of mass of the three points using two steps. First, we find the center of mass of A and B (Fig. 1.29).

$$(2m)(M) = (m)(A) + (m)(B)$$

We see the center of mass M lies at $(A + B)/2$. Moving points A and B to M , we have two points with masses m and $2m$ located at C and M respectively. Because CM connects a vertex to the midpoint of the opposite edge, the two points lie on the median. We now find the center of mass P of M and C .

$$(3m)(P) = (2m)(M) + (m)(C) = (2m)(A + B)/2 + (m)(C)$$

We find the center of mass P of the three points lies at $(A + B + C)/3$. Now, we can repeat this process two more times, beginning with different pairs of masses. Each time we show that the center of mass lies on each median, ending up with the exact same point. That means P is a point that lies on all three medians.

Recall that the three medians are only concurrent at a single point, the centroid. P happens to be a point which satisfies this constraint and also happens to be the average of the vertices. We finally see that the average of the vertices lies at the center of mass.

REMARK: In the above proof, we utilized capital letters as replacements for vectors. However, they are in essence the same.

1.3 N-Points Theorem Tetrahedron

Solution 1: We cut our tetrahedron into tetrahedron shells (akin to four triangles glued together at the edges). Using a similar argument as before, the individual centers of each shell will be the same as the center of mass of the entire tetrahedron. That means we can take the limit as the number of shells goes to infinity and only find the center of mass of the outermost shell.

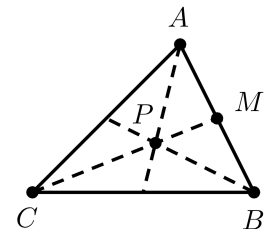


Figure 1.29

We can then imagine 4 points, each with mass m , located at each vertex. Notice that every vertex touches 3 faces. That tells us to imagine cutting each point into three pieces, each with mass $m/3$, such that each piece is reserved for one adjacent face. In the previous problem, we showed that three points located at each vertex contribute an equal amount of position-mass compared to an entire triangle. Hence, each triangular face on the shell is the equivalent of three pieces, located at each vertex.

Because there are three faces touching each vertex, each face contributes one piece, and three pieces make a point, we finally get back our original 4 points. We see that we can construct a tetrahedron shell from 4 points meaning the center of mass of our original tetrahedron also lies at the average of the four vertices.

REMARK: As before, despite the triangles having different areas, they have the same mass. This results from the differing widths of each face, a necessary condition for the slices to add back up the the whole.

Solution 2: We can instead cut the tetrahedron into triangular slices such that each slice is parallel to one of the faces (Fig. 1.30). We can then take the limit as the number of slices goes to infinity such that the center of mass of each slice lies exactly at the centroid. The resulting line of centers formed by all the slices forms a median; as before each median presents a valid method for finding the center of mass. Hence, the center of mass must lie at the intersection of all the medians, known as the centroid.

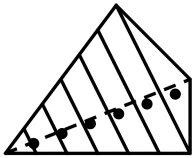


Figure 1.30

We propose the average of the vertices is a point that satisfies this constraint. To show this, we assume each vertex: A , B , C , and D , has a mass m . We will then find the center of mass in two steps. First, we find the center of mass M of A , B , and C .

$$(3m)(M) = (m)(A) + (m)(B) + (m)(C)$$

We find the center of mass M lies at $(A + B + C)/3$, the centroid of the outermost face. That means the line DM is a median, and the center of mass of all four vertices lies on it. Moving A , B , and C to M and finding the center of mass P of D and M , we see (Fig. 1.31)

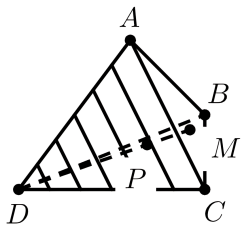


Figure 1.31

$$(4m)(P) = (3m)(M) + (m)(D) = (3m)(A + B + C)/3 + (m)(D)$$

The center of mass of all four points P lies at $(A + B + C + D)/4$. We can then repeat the process, each time beginning with a different face. We get the same result each time showing that P lies on all of the medians. Hence, the center of mass of a tetrahedron lies at the average of all the vertices.

REMARK: The logic follows exactly from the previously problem. Review that if you are confused.

1.4 Two Points

We can assign our origin to lie on m_1 . That means only m_2 contributes position mass $m_2 d$ to the system (Fig. 1.32). Using the discrete definition:

$$(m_1 + m_2)(r_1) = (m_2)(d)$$

Here, the distance of the center of mass C from the origin equals the distance of the center of mass from m_1 . We can then repeat this process placing our origin at m_2 .

$$(m_1 + m_2)(r_2) = (m_1)(d)$$

Solving each respectively, we find $r_1 = \frac{m_2 d}{m_1 + m_2}$ and $r_2 = \frac{m_1 d}{m_1 + m_2}$.

REMARK: Placing the center of mass at one of the points can greatly simplify the algebra.

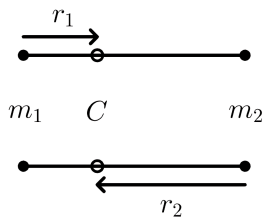
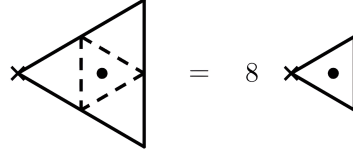


Figure 1.32

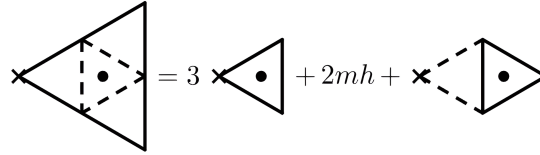
1.5 Equilateral Triangle

Solution 1: Let the triangle have a mass m and height h . Let all position-mass be calculated relative to the origin located at the X . We can create a system of equations utilizing scaling arguments, superposition, and a shift of origin.

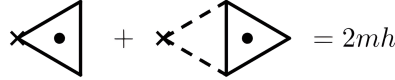
We can scale the equilateral triangle by a factor of 2 resulting in 4 times the mass and 2 times the length of the center of mass vector.



We can then break the scaled triangle into four parts. The top and bottom right corner essentially have an origin shifted h to the left.



Two triangles with total mass $2m$, attached base to base, have a center of mass exactly at the halfway point.



Solving this system of equations for the position-mass of a regular equilateral triangle $2mh/3$ and dividing by mass m , we find $r_{cm} = 2h/3$.

REMARK: Using $h = \sqrt{3}\ell/2$ gives $r_{cm} = \ell/\sqrt{3}$.

Solution 2: Let equilateral triangle S_1 be formed by the superposition of the smaller equilateral triangle S_2 and the trapezoid S_3 . We will then use a limiting argument to find the center of mass (Fig. 1.33).

We then let S_1 and S_2 have heights a and b . By superposition, we know $m_1 = m_2 + m_3$. We can then solve for the mass of each object using proportionality.

$$m_1 = m \quad m_2 = \frac{b^2}{a^2}m \quad m_3 = \left(1 - \frac{b^2}{a^2}\right)m$$

We let the center of mass vector for an equilateral triangle and trapezoid take the form αr and βr . We now find the center of mass vector for each object.

$$r_1 = \alpha a \quad r_2 = \alpha b \quad r_3 = \beta b$$

Finally, by superposition we know $\psi_1 = \psi_2 + \psi_3$. Solving for ψ_3 and taking the limit as $S_2 \rightarrow S_1$,

$$\lim_{b \rightarrow a} \left(1 - \frac{b^2}{a^2}\right)m * \beta b = \lim_{b \rightarrow a} \left[m * \alpha a - \frac{b^2}{a^2}m * \alpha b\right]$$

The rest is just algebra. We cancel the m 's, factor out the α on the right side, and group terms under a common denominator.

$$\lim_{b \rightarrow a} \beta b \left(\frac{a^2 - b^2}{a^2}\right) = \lim_{b \rightarrow a} \alpha \left(\frac{a^3 - b^3}{a^2}\right)$$

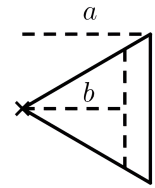


Figure 1.33

We then factor using the difference of squares and cubes, canceling like terms.

$$\begin{aligned}\lim_{b \rightarrow a} \beta b(a+b)(a-b) &= \lim_{b \rightarrow a} \alpha(a-b)(a^2 + ab + b^2) \\ \lim_{b \rightarrow a} \beta b(a+b) &= \lim_{b \rightarrow a} \alpha(a^2 + ab + b^2)\end{aligned}$$

Finally, we evaluate the limit.

$$\beta a(a+a) = \alpha(a^2 + a*a + a^2) \Rightarrow 2\beta = 3\alpha$$

In the limit, the trapezoid S_3 becomes a rod with center of mass located simply at $r_3 = \beta a = a$. Using $\beta = 1$, we find $\alpha = 2/3$. Hence, the center of mass of a triangle lies at $r_{cm} = 2h/3$.

REMARK: Notice we waited until the end to assume $\beta = 1$. The distance from the origin to the beginning of S_3 is b , but the center of mass r_3 of S_3 lies some distance further. Hence, $r_3 > b$. We defined the center of mass to be $r_3 = \beta b$, telling us $\beta > 1$ before we apply the limit. Only after the trapezoid becomes a rod and the sides and center of mass coincide with each other does β become one.

1.6 Regular Tetrahedron

Solution 1: Let the tetrahedron have a mass m and height h . Then, let all the position-mass be calculated relative to the origin located at one corner of the original tetrahedron. Before we begin however, we need to examine some geometric properties of regular tetrahedrons.

Any regular tetrahedron can be dissected into 4 smaller regular tetrahedrons T_s and a regular octahedron O , both with half the side length. Additionally, any regular octahedron O can be further dissected into 8 regular tetrahedrons T_m and 8 regular octahedrons with half the side length each.

The full tetrahedron dissection and the top half of the octahedron dissection are shown below. We wish to find the distance of the center of mass of the octahedron O (located at the geometric center) to the center of one face. We see that it happens to be the altitude of one of the mini-tetrahedrons T_m .

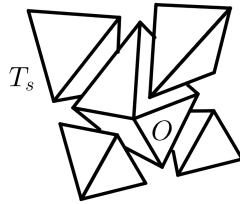


Figure 1.34

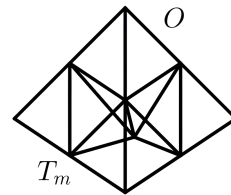


Figure 1.35

From visual inspection, we see that each mini-tetrahedron T_m has a side length half of the octahedron O . The octahedron shares a side length with the smaller regular tetrahedron T_s . If the smaller tetrahedron T_s has an altitude h , then by similar triangles, the mini-tetrahedron T_m has an altitude $h/2$.

Finally, we must notice that the mini-tetrahedron T_m and the smaller regular tetrahedron T_s are connected face to face. The combined heights $h + h/2$ connect one vertex of the original tetrahedron to the center of the octahedron. We'll see how this becomes useful in two ways.

Now, we are ready to construct our system of equations. Rather than use images (due to the difficulty in drawing), we'll let the larger tetrahedron, octahedron, and normal tetrahedron's have a position-mass ψ_L , ψ_O , and ψ .

We can scale the normal tetrahedron by a factor of 2 resulting in 8 times the mass and 2 times the length of the center of mass vector.

$$\psi_L = 16\psi$$

We can then break the scaled tetrahedron into five parts. The three corner tetrahedron's essentially have their origins shifted h to the left.

$$\psi_L = 4\psi + 3mh + \psi_O$$

Finally, we can condense the octahedron with mass $4m$ into a point located at the center of mass. That point is a distance $h + h/2$ away from the origin.

$$\psi_O = (4m)(3h/2)$$

Solving the system of equations for the position-mass of a regular tetrahedron $3mh/4$ and dividing by mass m , we find $r_{cm} = 3h/4$.

REMARK: <https://math.stackexchange.com/questions/3671664/splitting-the-octahedron> contains some nicer visuals. As a side note, once we had noticed that the octahedron's center of mass lied a distance $3h/2$ from one corner, we could've directly solved for the center of mass of the larger tetrahedron. The larger tetrahedron has a height $2h$ by proportionality and the octahedron's center of mass coincides with the larger tetrahedron's center of mass by symmetry. That means the larger tetrahedron's center of mass lies at $\frac{3}{4}(2h)$, or $3/4$ of the way up. However, we utilized a scaling argument for practice.

Solution 2: Let the regular tetrahedron S_1 be formed by the superposition of the smaller tetrahedron S_2 and the tetrahedron frustum S_3 . We will then reuse a limiting argument to find the center of mass (Fig. 1.36).

We then let S_1 and S_2 have heights a and b . By superposition, we know $m_1 = m_2 + m_3$. We again use proportionality to solve for the mass of each shape.

$$m_1 = \quad m_2 = \frac{b^3}{a^3}m \quad m_3 = \left(1 - \frac{b^3}{a^3}\right)m$$

We then let the center of mass vector for a regular tetrahedron and tetrahedron frustum take the form αr and βr . We now find each shapes center of mass.

$$r_1 = \alpha a \quad r_2 = \alpha b \quad r_3 = \beta b$$

Finally, by superposition about the same origin, we have $\psi_1 = \psi_2 + \psi_3$. Solving for ψ_3 as before and taking the same limit $S_2 \rightarrow S_1$,

$$\lim_{b \rightarrow a} \left(1 - \frac{b^3}{a^3}\right)m * \beta b = \lim_{b \rightarrow a} \left[m * \alpha a - \frac{b^3}{a^3}m * \alpha b\right]$$

Similar to last time, we just need to simplify from here. Rewriting each term with a common denominator, factoring out α , and removing mass m :

$$\lim_{a \rightarrow b} \beta b \left(\frac{a^3 - b^3}{a^3}\right) = \lim_{a \rightarrow b} \alpha \left(\frac{a^4 - b^4}{a^3}\right)$$

Next, we apply the difference of even powers and simplify.

$$\begin{aligned} \lim_{a \rightarrow b} \beta b(a - b)(a^2 + ab + b^2) &= \lim_{a \rightarrow b} \alpha(a^2 - b^2)(a^2 + b^2) \\ &= \lim_{a \rightarrow b} \alpha(a - b)(a + b)(a^2 + b^2) \\ \lim_{a \rightarrow b} \beta b(a^2 + ab + b^2) &= \lim_{a \rightarrow b} \alpha(a + b)(a^2 + b^2) \end{aligned}$$

Finally, we apply the limit.

$$\beta a(a^2 + a * a + a^2) = \alpha(a + a)(a^2 + a^2) \Rightarrow 3\beta = 4\alpha$$

In the limit, the tetrahedron frustum becomes a flat triangle with center of mass located simply at $r_3 = \beta a = a$. Using $\beta = 1$, we find $\alpha = 3/4$. Hence, the center of mass of a tetrahedron lies at $r_{cm} = 3h/4$.

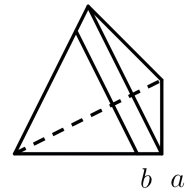


Figure 1.36

1.7 Half-Hoop

Solution 1: Usually when dealing with position-mass, we add the vectors by adding their components. Here, we will instead add them tail to tip directly. To do so, we will view a few case scenarios and generalize it to an continuous distribution.

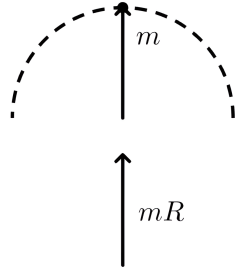


Figure 1.37

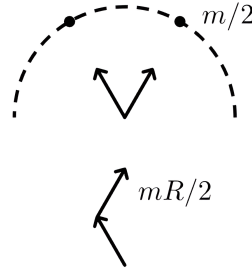


Figure 1.38

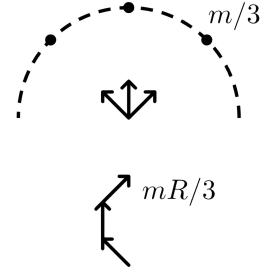


Figure 1.39

$N = 1$: Suppose that all the mass was concentrated at the middle of the hoop. The total position-mass is mR upwards.

$N = 2$: Suppose that all the mass was concentrated at two equally spaced points, each with mass $m/2$. Each point contributes a position-mass $mR/2$ towards it. Adding the vectors tail-to-tip, we find the total perimeter of the triangle-like shape to be $L = mR/2 + mR/2 = mR$.

$N = 3$: Suppose that all the mass was concentrated at three equally spaced points, each with mass $m/3$. Each point contributes a position-mass $mR/3$ towards it. Adding the vectors tail-to-tip, we find the total perimeter of the hexagon-like shape to be $L = mR/3 + mR/3 + mR/3 = mR$.

$N = n$: Suppose that all the mass was concentrated at n equally spaced points, each with mass m/n . Each point contributes a position-mass mR/n towards it. Adding the vectors tail-to-tip as always, we find the total perimeter to be $L = n * mR/n = mR$.

We see that approximating the hoop with more and more points doesn't change the length of the line formed by adding all the vectors tail to tip. Additionally, we see that approximating the hoop with more and more points results in a shape with more and more sides.

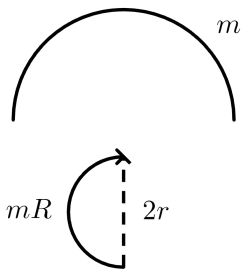


Figure 1.40

Finally, we take the limit as the number of points goes to infinity $n \rightarrow \infty$ meaning the points become a continuous hoop. Adding all the vectors tail-to-tip, their sum becomes a half-circle. We know the perimeter of the half-circle and the angle it subtends, meaning we can solve for the diameter using $\Delta s = r\Delta\theta$ (Fig. 1.40) .

$$\pi r = mR$$

We see that the total position-mass equals $\psi = 2r = 2mR/\pi$. We can now apply the center of mass definition.

$$mr_{cm} = 2mR/\pi$$

Finally, we see the center of mass lies at $r_{cm} = 2R/\pi$

Solution 2: Let the rod have a linear density λ . We can then cut the arc into $\Delta\theta$ slices and find the position-mass of each slice.

$$\Delta m = \lambda R \Delta\theta$$

$$r(\theta) = R \sin \theta$$

We find the position-mass of a slice to be $\Delta\psi = \lambda R^2 \sin\theta \Delta\theta$. Next, we can draw a right triangle located at the slice to relate Δx , $R\Delta\theta$, and θ using similar triangles (Fig. 1.41) .

$$\sin\theta = \Delta x / R\Delta\theta$$

Substituting, we find the position-mass of a slice to be $\Delta\psi = \lambda R \Delta x$. We see that any $\Delta\theta$ slice has an equivalent position-mass as a Δx slice located at $y = R$. The decrease in mass by a factor $\sin\theta$ is compensated by the increase in the position-vector by a factor $1/\sin\theta$.

Noticing this, we can project all the mass onto a horizontal line with length $2R$. We can then easily find the position-mass of the line segment ψ_L (Fig. 1.42) .

$$\psi_L = (\lambda * 2R)(R)$$

Because the position-mass of the line segment equals the position-mass of the entire hoop, we now know the total position-mass. We can then apply the definition of the center of mass.

$$(\lambda * \pi R)(r_{cm}) = (\lambda * 2R)(R)$$

Solving, we find the center of mass to be $r_{cm} = 2R/\pi$

REMARK: The part where we drew a right triangle on our slice and utilized similar triangles only works in the limit where the number of slices goes to infinity as the interior angles of the slices approach right angles.

1.8 Half-Circle

Solution 1: Using center of mass by parts, we can cut our half-circle into isosceles triangles and condense each into a point. Using the N -Points theorem, the center of mass of a triangle lies at $2h/3$ (Fig. 1.43) .

Similar to before, as the number of triangles goes to infinity, the approximation of the circle becomes exact. Then, the centers of all the triangles forms a hoop with radius $r = 2R/3$.

From the previous problem, the center of mass of a hoop lies at $r_{cm} = 2r/\pi$. Combining these results, we find the center of mass lies at $r_{cm} = 4R/3\pi$

REMARK: Although the mass of each triangular slice goes to zero and they become infinitely thin, the mass is still distributed with linearly increasing density. They do not become rectangles meaning the N -Points Theorem still applies.

Solution 2: We can instead use a limiting arguments taking the difference of two half-circles and limit as their difference approaches a hoop. Rather than jump right in, we notice that the mass of a circle is proportional to area. This is the same case as for a triangle. We can then reapply the same limiting case formula as before.

$$2\beta = 3\alpha$$

Recall that β was the coefficient of the limit. In this case, the center of mass of a hoop lies at $r = \beta R = (2/\pi)R$ meaning $\beta = 2/\pi$. We can then solve for α which we find to be $\alpha = 4/3\pi$. We then see that the center of mass lies at $r_{cm} = 4R/3\pi$

REMARK: If you don't see why we can do this, try deriving the limiting case formula as before. Think about why we get the same result and the physical significance of β and α .

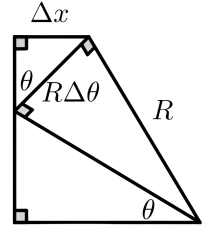


Figure 1.41

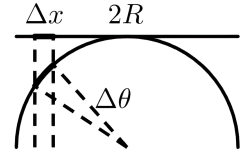


Figure 1.42

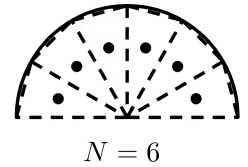


Figure 1.43

1.9 Arc

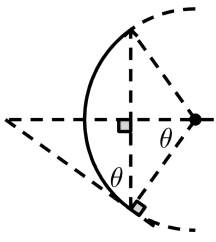


Figure 1.44

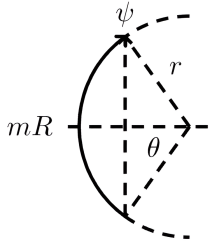


Figure 1.45

Solution 1: We can re-apply the circular vectors technique from before. However, when adding the vectors of each point tail-to-tip, we notice that the initial vector begins and ends at an angle θ to the vertical rather than completely horizontal. That means, the sum of all the vectors forms an arc rather than a half-circle (Fig. 1.44).

It may seem obvious that by symmetry, if the original arc subtends an angle 2θ , then the sum of the position-mass vectors also subtends the same angle. However, we can still draw similar triangles to confirm this fact (Fig. 1.45).

As before we know the length of the arc is given by mR . Then, we can solve for radius of the imaginary position-mass circle.

$$r * 2\theta = mR$$

We find $r = mR/2\theta$. The total position-mass is $\psi = 2r \sin \theta = mR \sin \theta / \theta$. Finally, we apply the center of mass definition.

$$mr_{cm} = mR \sin \theta / \theta$$

We solve to find $r_{cm} = R \sin \theta / \theta$

Solution 2: We assume the arc has a linear density λ and project all the mass onto a line with equivalent width located at $y = R$. The width of the line equals the width of the arc $w = 2R \sin \theta$, meaning we can find the position-mass of the line-segment.

$$\psi = (\lambda * 2R \sin \theta)(R)$$

We then can find the mass of the arc $m = \lambda * R * 2\theta$. Finally, we apply the center of mass definition.

$$(2R\lambda\theta)(r_{cm}) = (2R^2\lambda \sin \theta)$$

Solving for the center of mass, we find $r_{cm} = R \sin \theta / \theta$

REMARK: Revisit half-hoop if you are confused about the methods we used here.

1.10 Hollow Hemisphere

Solution 1: We claim that a hollow hemisphere is the equivalent of a vertical uniform density rod with length R . To show this, we can cut our hollow hemisphere into horizontal slices.

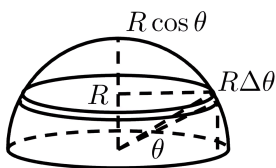


Figure 1.46

At angle θ and slice $\Delta\theta$, we can find the mass by treating the slice as a hoop with thickness equal to the arc length (Fig. 1.46).

$$\Delta m = \sigma(2\pi R \cos \theta)(R \Delta \theta)$$

Then, we can draw a right triangle onto our slice and use similar triangles to relate Δy , $R \Delta \theta$, and θ (Fig. 1.47).

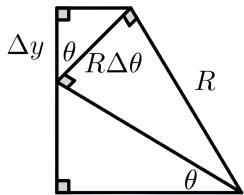


Figure 1.47

$$\cos \theta = \Delta y / R \Delta \theta$$

Substituting, we find $\Delta m = 2\pi\sigma R \Delta y = k \Delta y$. We see that at any horizontal slice, the mass is constant. Hence, the center of mass lies halfway in between, located at

$$r_{cm} = R/2$$

REMARK: The reason for the constant mass lies in the competing effects of decreasing radius and increasing curvature. At any given slice, the radius shrinks by a factor $\cos \theta$ while the curvature increases by a factor $1/\cos \theta$, canceling each other out. This reasoning is the basis of the formula for the area of a sphere. We can project the sphere onto a cylinder because

each slice has the same mass (meaning the same area). The cylinder then has a circumference $2\pi R$ and a height $2R$, giving us the surface area $A = 4\pi R^2$.

Solution 2: Let a horizontal slice have a circumference C making the position-mass of the slice:

$$\Delta\psi = \Delta m * r(\theta) = (\sigma * C * R\Delta\theta)(R\sin\theta)$$

In Half-Hoop, we showed that using similar triangles on $\Delta\theta$ slice, $\sin\theta = \Delta x/R\Delta\theta$. Substituting, we find we find $\Delta\psi = \sigma C\Delta x * R$.

That is the equivalent of a flat circular slice located at $y = R$. We can then project all the slices forming a flat circle. We now calculate the total position-mass.

$$\psi = (\sigma * \pi R^2)(R)$$

The circular projection (with radius R) has the same position-mass as the original hemisphere. We finally apply the center of mass definition:

$$(\sigma * 2\pi R^2)(r_{cm}) = (\sigma * \pi R^2)(R)$$

Solving, we find $r_{cm} = R/2$.

1.11 Solid Hemisphere

Solution 1: When we approached half circle, we cut it into triangular slices and reduced it to a hoop. Similarly, we can cut the solid hemisphere into inverted-pyramid like slices, such that the apexes are all concurrent at the center of the sphere.

The N -Points theorem tells us the center of mass of each slices lies at $3h/4$. We can then take the limit as the number of slices goes to infinity. The centers of all the pyramids form a hollow hemisphere with radius $3R/4$.

We can apply the previous problems result to find the center of mass. Doing so, we see $r_{cm} = 3R/8$.

REMARK: We could've instead bounded the solid hemisphere between a cylinder with the center of mass located at $R/2$ and a solid cone with the center of mass located at $R/4$. The average of the bounds happens to be exactly equal to the actual center of mass.

Solution 2: We can reuse the same limiting coefficient relationship for two solids which we derived in regular tetrahedron.

$$3\beta = 4\alpha$$

In the limit, the hollow hemisphere has a center of mass located at $r = \beta R = R/2$ meaning $\beta = 1/2$. We then solve for α to find $\alpha = 3/8$. We see the center of mass lies at $r_{cm} = 3R/8$

REMARK: Both a tetrahedron and a solid hemisphere have a mass proportional to volume or length cubed. This results in the same powers affecting the limits, meaning it all mathematically works out the same. You can re-derive the limiting relationship if you aren't convinced.

1.12 Trapezoid

We can superpose a smaller triangle with height h' to complete the isosceles trapezoid S_1 , forming an isosceles triangle S_2 . We can then superpose a negative smaller triangle S_3 to return the system back to its original state.

Now we have two triangles with heights $h + h'$ and h' . Letting the larger triangle have a mass $m_2 = m$, we can find the mass of the smaller triangle and the original trapezoid using proportionality and the superposition relationship $m_1 = m_2 + m_3$.

$$m_3 = -\frac{a^2}{b^2}m$$

$$m_1 = \left(1 - \frac{a^2}{b^2}\right)m$$

Then, using the N -Points theorem, we can find the center of mass of the smaller triangle and the larger triangle noting that the base of the smaller triangle lies a distance h above the origin.

$$r_2 = (h + h')/3$$

$$r_3 = h + h'/3$$

Finally, we can relate the heights and bases of the triangles using similar triangles (Fig. 1.48) .

$$\frac{h'}{h + h'} = \frac{a}{b}$$

Solving for h' , we find $h' = ah/(b - a)$. Now, we can apply center of mass by parts.

$$\left(\frac{b^2 - a^2}{b^2}m\right)(r_{cm}) = (m)(h + h')/3 + \left(-\frac{a^2}{b^2}m\right)(h + h'/3)$$

We don't want h' to be in our answer, meaning we have to substitute using our similar triangles relationship. The rest is just algebra.

$$\begin{aligned} \frac{b^2 - a^2}{b^2}r_{cm} &= \frac{hb}{3(b - a)} - \frac{a^2}{b^2}\left(h + \frac{ah}{3(b - a)}\right) \\ \frac{(b - a)(b + a)}{b^2}r_{cm} &= \frac{h}{3(b - a)b^2} [b^3 - 3a^2(b - a) - a^3] \\ (b - a)^2(b + a)r_{cm} &= (h/3)(b^3 - 3ba^2 + 2a^3) \end{aligned}$$

Using the factor theorem for multi-variable polynomials, we see $b = a$ is a root to the right side polynomial (which also happens to be a double root). Using synthetic division, we can factor the cubic and the resulting quadratic.

$$(b - a)^2(b + a)r_{cm} = (h/3)(b - a)^2(2a + b)$$

Finally, we can solve to see $r_{cm} = h(2a + b)/3(a + b)$

REMARK: We can alternatively use center of mass by parts directly, cutting the trapezoid into the equivalent of a rectangle and a triangle, to find the center of mass.

1.13 Conical Frustum

In the similar spirit of Trapezoid, we superpose a smaller cone on top of the conical frustum S_1 to create a large cone S_2 . We then superpose a negative smaller cone S_3 to return the system to its original state.

Let the larger cone S_2 have a mass m . We can then find the masses of the other two objects using proportionality and superposition noting $m_1 = m_2 + m_3$.

$$m_3 = -\frac{a^3}{b^3}m$$

$$m_1 = \left(1 - \frac{a^3}{b^3}\right)m$$

We again reuse the N -Points Theorem noting the smaller cone's base lies a distance h above the origin.

$$r_2 = (h + h')/4$$

$$r_3 = h + h'/4$$

Finally, we can reuse the same similar triangle relationship as before (Fig. 1.49) .

$$\frac{h'}{h + h'} = \frac{a}{b}$$

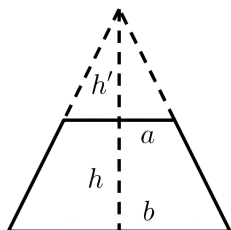


Figure 1.48

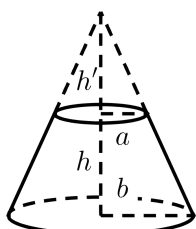


Figure 1.49

Now, we can apply the center of mass definition, substituting $h' = ah/(b-a)$ and simplifying.

$$\begin{aligned} \left(\frac{b^3 - a^3}{b^3}m\right)(r_{cm}) &= (m)(h+h')/4 + \left(-\frac{a^3}{b^3}m\right)(h+h'/4) \\ \frac{b^3 - a^3}{b^3}r_{cm} &= \frac{hb}{4(b-a)} - \frac{a^3}{b^3}\left(h + \frac{ah}{4(b-a)}\right) \\ \frac{(b-a)(b^2 + ab + a^2)}{b^3}r_{cm} &= \frac{h}{4b^3(b-a)}(b^4 - 4a^3(b-a) - a^4) \\ (b-a)^2(b^2 + ab + a^2)r_{cm} &= (h/4)(b^4 - 4ba^3 + 3a^4) \end{aligned}$$

As before, we quickly see $b = a$ is a root using the factor theorem for multi-variable polynomials. We can then factor the right hand side using synthetic division twice.

$$(b-a)^2(b^2 + ab + a^2)r_{cm} = (h/4)(b-a)^2(b^2 + 2ab + 3a^2)$$

Solving, we find $\boxed{r_{cm} = h(3a^2 + 2ab + b^2)/4(a^2 + ab + b^2)}$

Chapter 2

Moment of Inertia

2.1 Solutions to Exercises

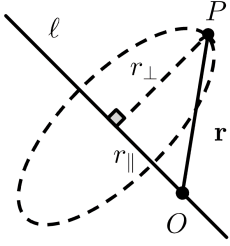


Figure 2.1

Q2.1

When an point P goes in a circle around the axis ℓ , the radius of the circle is the perpendicular distance. Relative to an arbitrary origin O located on ℓ , the position vector of P has no physical significance.

REMARK: Given that the origin O lies on the axis of rotation ℓ , we can first find the parallel distance along the axis using the position vector $\mathbf{r}$ of a point P and the direction vector $\hat{\ell}$ of the line.

$$r_{\parallel} = \mathbf{r} \cdot \hat{\ell}$$

Then, we can find the perpendicular distance using the Pythagorean theorem (Fig. 2.1) .

$$r_{\perp}^2 + r_{\parallel}^2 = r^2$$

We finally remember that the dot product of the position vector with itself equals the magnitude squared or $r^2 = \mathbf{r} \cdot \mathbf{r}$, allowing us to finally solve for the perpendicular distance.

Q2.2

Consider a system S with $N = 2$ point masses, m_1 and m_2 , located at r_1 and r_2 relative to the axis of rotation ℓ . We can then find the total moment of inertia:

$$I = m_1 r_1^2 + m_2 r_2^2$$

We can generalize the above result to a system with N point masses:

$$I = m_1 r_1^2 + m_2 r_2^2 + \dots + m_N r_N^2$$

Finally, we can condense the above summation to write $I = \sum_{k=1}^N m_k r_k^2$

Q2.3

We let the object S be cut into slices, each with mass Δm at a radial distance r . Then, we find the total moment of inertia to be:

$$I = \sum \Delta m r^2$$

We then take the limit as the mass of each slice goes to zero meaning $\Delta m \rightarrow dm$.

$$I = \lim_{\Delta m \rightarrow 0} \sum \Delta m r^2$$

In the limit, the resulting summation becomes the integral $I = \int_S r^2 dm$

Q2.4

Recall that the associative property of addition says that grouping terms in parenthesis doesn't affect the final sum:

$$(a + b) + c = a + (b + c)$$

This means we can group the moment of inertia of multiple points into a subsystem.

$$I = (m_1 r_1^2 + \dots + m_i r_i^2) + (m_{i+1} r_{i+1}^2 + \dots + m_n r_n^2)$$

Defining the first set as S_1 and the second set as S_2 such that $S_1 + S_2 \equiv S$, we see

$$I = I_1 + I_2$$

regardless of how we choose to split the original system S .

Q2.5

Suppose N points in a system S are all located a distance r directly away from the axis of rotation. The total moment of inertia of the system would then be:

$$I = \sum_{k=1}^N m_i r^2$$

Now, we define a new point with total mass equal to the sum of all N points.

$$M = \sum_{k=1}^N m_i$$

We can then rewrite the total moment of inertia in terms of M noting r is constant.

$$I = \sum_{k=1}^N m_i r^2 = r^2 \sum_{k=1}^N m_i = M r^2$$

We see that we can superpose all N points without changing the moment of inertia.

REMARK: Supposing that we did superpose all N points into one location, the mass density at that location would increase. Although the system would change, the moment of inertia wouldn't however.

Q2.6

We approach this in the same manner as before. Assume a system S contains N planar slices, all with mass Δm and major length ℓ . We find the total moment of inertia to be:

$$I = \sum_{k=1}^N \beta \Delta m \ell^2$$

We now define a single plane with total mass equal to the sum of all N planes.

$$M = \sum_{k=1}^N \Delta m$$

We can then rewrite the total moment of inertia in terms of M , noting ℓ and β are constant.

$$I = \sum_{k=1}^N \beta \Delta m \ell^2 = \beta \ell^2 \sum_{k=1}^N \Delta m = \beta M \ell^2$$

We see that we can superpose all N planes without changing the moment of inertia.

REMARK: Let's consider the more general case of an object S that is cut into slices such that each planar slice is perpendicular to the axis of rotation (Fig. 2.2). We do not assume that each slice is symmetrical, but we still superpose them such that all the slices lie on the same plane. Only the vertical position of each point changes (relative to the axis of rotation), meaning the total moment of inertia doesn't change. Recall that only the perpendicular radial distance matters. Make sure you understand why this more general technique, known as squashing, follows from superposition.

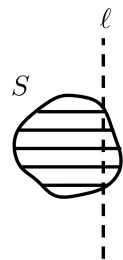


Figure 2.2

Q2.7

Both the object S and the hoop H have a mass m . The moment of inertia of the object is given by $I_S = \beta md^2$. We can then find the moment of inertia of the hoop.

$$I_H = m \left(d\sqrt{\beta} \right)^2 = \beta md^2$$

We see the hoop and object have the same moment of inertia.

REMARK: We see that the radius of gyration is the rotational analog of the center of mass. However, we place greater emphasis on the total moment of inertia rather than the radius of gyration, unlike before where the center of mass held greater importance than the total position-mass.

E2.8

We apply the discrete definition of moment of inertia.

$$I = \sum m_i r_i^2 = md^2 + md^2 + md^2 + md^2$$

We find the total moment of inertia of the system to be $I = 4md^2$

REMARK: Notice that we used the perpendicular distance from the y-axis, not the radial distance from the origin. Additionally, the heights of the masses were irrelevant.

E2.9

Relative to the cylindrical axis, every point on the hoop is the same distance away. By superposition, we can superpose all the points into one point with equivalent mass. We know the definition of the moment of inertia of a point. Hence, the moment of inertia of the hoop is $I = mr^2$

E2.10

Recall that the vertical position doesn't affect the moment of inertia of an object. Hence, we can squash the cylinder into a disk such that the disk lies on a plane perpendicular to the axis of rotation. We are given the moment of inertia of a disk. Hence, we find the moment of inertia of the cylinder becomes $I = \frac{1}{2}mr^2$

E2.11

Again, the vertical position doesn't affect the moment of inertia. We can then squash the cone into a circle. Hence, the moment of inertia of a cone about its cylindrical axis is given by $I = \frac{1}{2}mr^2$.

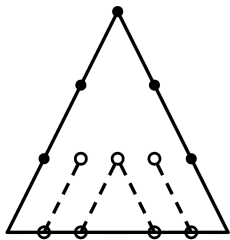


Figure 2.3

REMARK: Although the cone's surface lies at a slant relative to the plane to which it is squashed, the density of the resulting circle remains uniform. We can best see this from a side view, where we see that only the thickness of the resulting cylinder is affected by the slant. In (Fig. 2.3), the dark circles represent the original positions of each point and the open circles represent the ending positions of each point after being squashed. The dashed lines which represent the surface of the cone line up with the same height, with a finite thickness. But then we can squash it again, repeating the process. We see the density eventually becomes uniform.

E2.12

The i th point has a mass $m_i = m/N$ because all the points are equally sized and a distance from the origin $r_i = id$ because they are equally spaced. Using the discrete definition,

$$I = \sum_{i=1}^N m_i r_i^2 = \sum_{i=1}^N (m/N)(id)^2 = \frac{md^2}{N} \sum_{i=1}^N i^2 = \frac{md^2}{N} * \frac{N(N+1)(2N+1)}{6}$$

We find the total moment of inertia to be $I = \frac{md^2}{6}(N+1)(2N+1)$

REMARK: We can find the moment of inertia of a continuous rod by letting the furthest point be located at a fixed distance $\ell = Nd$ and taking the limit as the number of points goes to infinity $N \rightarrow \infty$ while the distance between each point goes to zero $d \ll \ell$.

$$\lim_{N \rightarrow \infty} I = \lim_{N \rightarrow \infty} \frac{md^2}{6}(N+1)(2N+1) = \lim_{N \rightarrow \infty} \frac{m\ell^2}{6}(1+1/N)(2+1/N) = \frac{1}{3}m\ell^2$$

Notice the moment of inertia decreases as the line of points becomes continuous. As every point splits to fill in the space behind it, they all move closer to the axis of rotation, decreasing the total moment of inertia. As a side note, if the axis of rotation was the x -axis, the moment of inertia would simply be zero because all the points lie on the axis. It's fair to always assume the axis of rotation is perpendicular to the object.

Q2.13

Recall that in an n dimensional object S , the size of each slice S_i scales as:

$$V \propto k^n$$

and the density of each slice given a density function $\rho(x) = x^s$ scales according to

$$\rho \propto k^s$$

As before, we can combine these two results while defining a new variable $r = n + s$ to find the new mass of each slice.

$$m_i \propto k^{n+s} = k^r$$

We then find the new moment of inertia by adding the moment of inertia of each slice.

$$I_k = \sum m'_i r_i'^2 = \sum m_i k^r * r_i^2 k^2 = k^{r+2} \sum m_i r_i^2$$

Substituting our initial moment of inertia, we see $I_k = I_0 k^{r+2}$.

REMARK: Notice the extra 2 which comes from moment of inertia being proportional to the radial distance squared. As before, do not try to memorize this formula. Simply consider three factors: size, density, and the radial distance of each slice. Consider how they change to find the resulting moment of inertia. Unlike position-mass where scaling occurs relative to the origin O , moment of inertia can be scaled relative to an origin O or an axis ℓ . Different methods of scalings will cause different changes.

Q2.14

We split an object S_1 into S_2 and S_3 . By superposition, we have:

$$I_1 = I_2 + I_3$$

This holds true regardless of the relative sizes of S_2 and S_3 . That means we can take the limit without affecting the equality.

$$\lim_{S_2 \rightarrow S_1} I_3 = \lim_{S_2 \rightarrow S_1} I_1 - I_2$$

As before, when S_2 becomes S_1 , S_3 becomes nothing. Hence, we get $0 = 0$.

REMARK: This idea parallels center of mass, where we cleverly avoided $0 = 0$ through some algebra. As before, moment of inertia is commonly written in the form $\beta m \ell^2$ where β is a numeric coefficient tied to the geometry of the shape. The shell and the solid have different numeric coefficients, meaning we can then establish a relationship between them. Knowing one allows us to solve for the other.

Q2.15

Consider a point on the line ℓ . It has no moment of inertia because its radial distance is zero. We can then find the moment of inertia of a point located relative to a different, parallel axis ℓ' , given that the distance between the axes is d .

$$I = I_{cm} + md^2 = 0 + md^2$$

We correctly get the moment of inertia of a point $I = md^2$.

Q2.16

We take the derivative of the moment of inertia of a system and set it equal to zero.

$$I' = \nabla_{\mathbf{r}} \int_S r^2 dm = 2 \int_S \mathbf{r} dm = 0$$

We see that the moment of inertia has a critical point at the center of mass. To determine whether it is a local minimum or maximum, we find the sign of the second derivative.

$$I'' = \nabla_{\mathbf{r}} \cdot \int_S \mathbf{r} dm = \int_S dm = M > 0$$

We see that moment of inertia has a local minimum when the axis of rotation crosses the center of mass.

REMARK: We can instead utilize the trivial inequality (a number squared is always positive):

$$d^2 \geq 0$$

We then multiply both sides by mass m and add I_{cm} .

$$I_{cm} + md^2 \geq I_{cm}$$

Equality is achieved when $d = 0$, or when the axis crosses the center of mass.

Q2.17

We utilize the Parallel Axis Theorem. The new axis of rotation is located a distance $\ell/2$ from the center of mass.

$$I = I_{cm} + md^2 = \frac{1}{12}m\ell^2 + m(\ell/2)^2 = \left(\frac{1}{12} + \frac{1}{4}\right)m\ell^2$$

We see that the moment of inertia of a rod hinged at one end is $I = \frac{1}{3}m\ell^2$.

E2.18

A rod with length ℓ hinged at its center of mass is the superposition of two rods with length $\ell/2$ hinged at their ends, not two rods with length ℓ . We can redefine the major length ℓ of a rod hinged at its center as $L = \ell/2$.

$$I = \frac{1}{12}m\ell^2 = \frac{1}{3}m(\ell/2)^2 = \frac{1}{3}mL^2$$

We see that a rod hinged at its center is indeed the superposition of two rods with half the length hinged at their ends.

REMARK: The reason for the small coefficient $\beta = 1/12$ when hinged at its center lies in the major length being twice the furthest distance any one point is. While such a choice may seem contradictory, it would also be un-intuitive to use the half-length of a rod.

Q2.19

We first consider a point P_1 with mass m on the x -axis located at $x = d$.

$$I_x = 0 \quad I_y = md^2 \quad I_z = md^2$$

We then consider a point P_2 with mass m on the y -axis located at $y = d$.

$$I_x = md^2 \quad I_y = 0 \quad I_z = md^2$$

We next consider a point P_3 with mass m on the xy -plane located at (d, d) .

$$I_x = md^2 \quad I_y = md^2 \quad I_z = m(d\sqrt{2})^2$$

For all three points, we see that $I_z = I_x + I_y$ (Fig. 2.4). We then test a point P_4 located on the z -axis at $z = d$.

$$I_x = md^2 \quad I_y = md^2 \quad I_z = 0$$

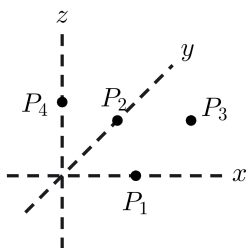


Figure 2.4

We see that the perpendicular axis theorem no longer holds

REMARK: If the perpendicular axis theorem doesn't apply to a point not located on the plane, then it certainly doesn't apply to a 3D object.

Q2.20

Consider an object S that exists in the xy -plane and a point O at which all three axes are concurrent. Notice that the perpendicular distance r_z from the z -axis of an arbitrary point P equals the radial distance r from the origin P (Fig. 2.5) .

$$r_z = r_O$$

We can then square both sides and multiply by the mass dm of that slice. We can then add the quantity up for the entire object S .

$$\int_S r_z^2 dm = \int_S r_O^2 dm$$

The left hand side is the moment of inertia I_z relative to the z -axis. We can then substitute this into the three-axes theorem.

$$I_x + I_y + I_z = 2 \int_S r_O^2 dm = 2I_z$$

We see by subtracting I_z from both sides, we get the perpendicular axis theorem.

E2.21

Let the cylindrical axis of the hoop be the z -axis such that the x and y axes lie along the diameter, meaning they are diametric axes (Fig. 2.6) . We can then apply the perpendicular axis theorem as a hoop is a planar object.

$$I_z = I_x + I_y$$

We know the moment of inertia of a hoop about the diametric axis: $I = mr^2$. We then note that I_x and I_y are the same because both axes which they are relative to lie along its diameter and the resulting figures are congruent under rotation.

$$I_x = I_y \equiv I_d$$

We can then solve for the moment of inertia about the diametric axis. We find it to be $I_d = \frac{1}{2}mr^2$

REMARK: We can repeat this process for a solid circle with moment of inertia $I = \frac{1}{2}mr^2$. Take a second to convince yourself that the moment of inertia relative to a diametric axis would be $I_d = \frac{1}{4}mr^2$ for a solid circle.

E2.22

The x and y position of a point r_{ix} and r_{iy} are scaled by a factor a/r and b/r respectively. We can then relate a points new position to its old one using a scaling argument.

$$r'_{ix} = \frac{a}{r} r_{ix} \quad r'_{iy} = \frac{b}{r} r_{iy}$$

We already solved for the moment of inertia of a hoop relative to a diametric axis, which we can express using the discrete definition.

$$\sum m_i r_{ix}^2 = \sum m_i r_{iy}^2 = \frac{1}{2}mr^2$$

We can then solve for the new moment of inertia relative to each diametric axis using a scaling argument. Recall that the circle is stretched to form an ellipse, meaning the mass of each point doesn't change. Relative to the major axis:

$$I_x = \sum m_i r_{iy}^2 = \sum m_i \left(\frac{b}{r} r_{iy} \right)^2 = \left(\frac{b}{r} \right)^2 \sum m_i r_{iy}^2 = \left(\frac{b}{r} \right)^2 \left(\frac{1}{2}mr^2 \right)$$

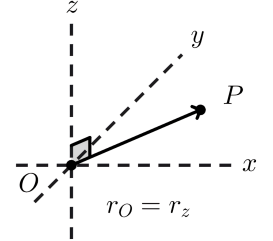


Figure 2.5

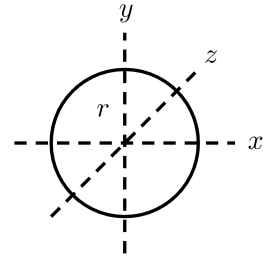


Figure 2.6

We can repeat the process with the minor axis to find:

$$I_x = \frac{1}{2}mb^2 \qquad I_y = \frac{1}{2}ma^2$$

Finally, we use the perpendicular axis to find the moment of inertia relative to the cylindrical axis of the ellipse.

$$I_z = I_x + I_y = \frac{1}{2}mb^2 + \frac{1}{2}ma^2$$

We find the total moment of inertia of the ellipse to be $I = \frac{m}{2}(a^2 + b^2)$

REMARK: Make sure you can go through the reasoning in your head. All of it follows from simple proportionality. We stretch a circle, meaning each moment of inertia component also stretches by the same factor squared. We can then recombine the components using the perpendicular axis theorem. Compared to integration, there's relatively minimal mathematical content in the above derivation. Think about that we labeled the distance from the x -axis as r_{iy} . Additionally, we can check the limit in which $a = b \equiv r$; the moment of inertia correctly reduces to $I = mr^2$ for a hoop as we would expect. Finally, we can repeat the above derivation for a solid ellipse. Convince yourself the total moment of inertia would be $I = \frac{m}{4}(a^2 + b^2)$ given that the moment of inertia of a solid circle is $I = \frac{1}{2}mr^2$ about its cylindrical axis.

E2.23

We can squash the sphere about its z axis to form a circle (with non-uniform density). As the moment of inertia doesn't change from the original $I = \frac{2}{3}mr^2$, we can apply the perpendicular axis theorem alongside radial symmetry to find the moment of inertia about a diametric axis I_d .

$$I_d = \frac{1}{3}mr^2$$

Then, we can stretch the circle relative to major and minor axis to form an ellipse. Recall that stretching by a factor k scales the moment of inertia by a factor k^2 .

$$I_x = \frac{1}{3}ma^2 \qquad I_y = \frac{1}{3}mb^2$$

Finally, we combine the moment of inertia using the perpendicular axis theorem, finding the moment of inertia relative to the z -axis. Un-squashing our ellipse to get our ellipsoid, we find the total moment of inertia to be $I = \frac{m}{3}(a^2 + b^2)$

REMARK: The majority of the reasoning follows from the previous exercises. Review them if you are confused. We can repeat this process for a solid ellipsoid by beginning with the moment of inertia of a solid sphere $I = \frac{2}{3}mr^2$. Convince yourself that a solid ellipsoid's moment of inertia about its z -axis would be $I = \frac{m}{5}(a^2 + b^2)$. Additionally, we can check the limit of $a = b \equiv r$; for a hollow ellipsoid, it correctly reduces to $I = \frac{2}{3}mr^2$, the moment of inertia of a hollow sphere.

2.2 Solutions to Problems

2.1 Parallel Axis Theorem

Consider a point with mass m located a distance r away from the axis of rotation. We know it has a moment of inertia $I = mr^2$. Now, we consider what happens when the mass splits into two parts, each with mass $m/2$ that travel in opposite directions a distance ϵ (Fig. 2.7).

Case 1: We first consider the case where they travel radially towards and away from the axis of rotation. We can recalculate the new moment of inertia using the discrete

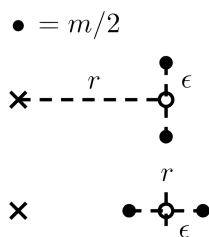


Figure 2.7

definition.

$$\begin{aligned} I_r &= (m/2)(r - \epsilon)^2 + (m/2)(r + \epsilon)^2 \\ &= \frac{m}{2} [(r^2 - 2r\epsilon + \epsilon^2) + (r^2 + 2r\epsilon + \epsilon^2)] \\ &= mr^2 + m\epsilon^2 \end{aligned}$$

Case 2: We then consider the case where they travel tangentially away from their initial position. We again recalculate the new moment of inertia using the Pythagorean theorem.

$$\begin{aligned} I_t &= (m/2)(r^2 + \epsilon^2) + (m/2)(r^2 + \epsilon^2) \\ &= mr^2 + m\epsilon^2 \end{aligned}$$

In case 1, the terms linear in ϵ canceled while in both scenarios, the quadratic terms added. The moment of inertia ends up being the sum of mr^2 – the moment of inertia if all the mass was concentrated at the center of mass – and $m\epsilon^2$ – the moment of inertia of the all the mass relative to an axis through the center of mass.

Finally, we assume a point mass m begins a distance d away from the origin. It then ejects mass arbitrarily in all directions such that the center of mass doesn't change. We have shown that all the linear differences cancel around the center of mass while the quadratic terms add up, meaning the resulting moment of inertia becomes $I = md^2 + I_{cm}$

2.2 Perpendicular Axis Theorem

Review the calculus solution. Just replace all the integrals with summations.

2.3 Three Axes Theorem

Again, review the calculus solution. Simply replace all the integrals with summations.

2.4 Rod

Solution 1: We can create a system of equations using scaling arguments, superposition, and the parallel axis theorem. Let the X denote the axis of rotation.

We can scale the rod by a factor of 2 resulting 2 times the mass and 4 times the average distance squared.

$$\text{---} \times \text{---} = 8 \quad \text{---} \times \text{---}$$

By superposition, the scaled rod consists of two normal rods hinged at their ends.

$$\text{---} \times \text{---} = 2 \quad \times \text{---}$$

Lastly, by the parallel axis theorem, we can relate the moment of inertia of a rod hinged at one end to that hinged at its center.

$$\text{---} \times = \text{---} \times + \frac{1}{4} m \ell^2$$

Solving the system of equations, we find $I_{cm} = \frac{1}{12} m \ell^2$

Solution 2: We will construct a limiting argument with a larger rod S_1 with length a , a smaller rod S_2 with length b , and their difference S_3 (Fig. 2.8) .



Figure 2.8

Assuming the larger rod has a mass m , we can find the masses of the other two rods using proportionality and superposition $m_1 = m_2 + m_3$.

$$m_1 = m \quad m_2 = \frac{b}{a}m \quad m_3 = \left(1 - \frac{b}{a}\right)m$$

Then, we can assume the moment of inertia of S_1 and S_2 take the form $I = \alpha m \ell^2$ while the moment of inertia of S_3 takes the form $I = \beta m \ell^2$.

Finally, we can apply the superposition relationship $I_1 = I_2 + I_3$, solve for I_3 and take the limit as $S_2 \rightarrow S_1$.

$$\lim_{b \rightarrow a} \beta \left(1 - \frac{b}{a}\right) m b^2 = \lim_{b \rightarrow a} \left[\alpha m * a^2 - \alpha \frac{b}{a} m * b^2 \right]$$

As always, we don't evaluate the limit right away, we first cancel common terms, factor, and simplify.

$$\begin{aligned} \lim_{b \rightarrow a} \beta \left(\frac{a-b}{a}\right) b^2 &= \lim_{b \rightarrow a} \alpha \left(\frac{a^3 - b^3}{a}\right) \\ \lim_{b \rightarrow a} \beta (a-b) b^2 &= \lim_{b \rightarrow a} \alpha (a-b)(a^2 + ab + b^2) \\ \lim_{b \rightarrow a} \beta b^2 &= \lim_{b \rightarrow a} \alpha (a^2 + ab + b^2) \end{aligned}$$

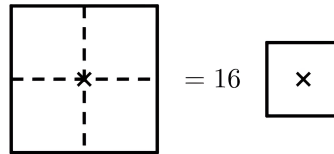
Finally, we apply the limit to find

$$\beta(a^2) = \alpha(a^2 + a * a + a^2) \Rightarrow \beta = 3\alpha$$

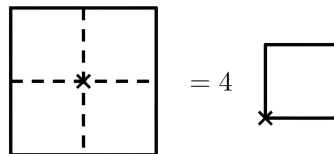
In the limit S_3 becomes a point with moment of inertia $I = \beta m_3 a^2 = m_3 a^2$. We see $\beta = 1$ meaning $\alpha = 1/3$. So, for a rod hinged at one end, we find $I = \frac{1}{3} m \ell^2$

2.5 Square

Solution 1: We can scale the square by a factor of 2, resulting 4 times the mass and 4 times the average square distance.



By superposition, the scaled square consists of 4 normal squares hinged at their corners.



Lastly, by the parallel axis theorem, we can relate the moment of inertia of a square hinged at one end to that hinged at its center.

$$\text{Square hinged at corner} = \text{Square hinged at center} + \frac{1}{2} m \ell^2$$

Solving the system of equations, we find $I_{cm} = \frac{1}{6} m \ell^2$.

Solution 2: We cut a larger square S_1 with side length a into a smaller square S_2 with side length b and a shell S_3 . We then find their masses using superposition (Fig. 2.9).

$$m_1 = m \quad m_2 = \frac{b^2}{a^2}m \quad m_3 = \left(1 - \frac{b^2}{a^2}\right)m$$

Next, we assume a square has a moment of inertia $I = \beta m \ell^2$. In the limit as the shell becomes 4 rods (which we can superpose into one denser rod). We can use the Parallel Axis Theorem to find the resulting coefficient.

$$I = md^2 + I_{cm} = m(\ell/2)^2 + \frac{1}{12}m\ell^2 = \frac{1}{3}m\ell^2$$

We see $\beta = 1/3$ in the limit. Finally, we can apply the superposition relationship $I_1 = I_2 + I_3$, solving for I_3 and taking the limit as $S_2 \rightarrow S_1$.

$$\lim_{b \rightarrow a} \beta \left(\frac{b^2 - a^2}{a^2} \right) mb^2 = \lim_{b \rightarrow a} \left[\alpha ma^2 - \alpha \left(\frac{b^2}{a^2} \right) mb^2 \right]$$

The rest is just algebra.

$$\begin{aligned} \lim_{b \rightarrow a} \beta b^2 \left(\frac{b^2 - a^2}{a^2} \right) &= \lim_{b \rightarrow a} \alpha \left(\frac{a^4 - b^4}{a^2} \right) \\ \lim_{b \rightarrow a} \beta b^2 (b - a)(b + a) &= \lim_{b \rightarrow a} \alpha (a^2 + b^2)(a^2 - b^2) \\ &= \lim_{b \rightarrow a} \alpha (a^2 + b^2)(a + b)(a - b) \\ \lim_{b \rightarrow a} \beta b^2 (b + a) &= \lim_{b \rightarrow a} \alpha (a^2 + b^2)(a + b) \end{aligned}$$

Finally, we evaluate the limit.

$$\beta a^2(a + a) = \alpha(a^2 + a^2)(a + a) \Rightarrow \beta = 2\alpha$$

Previously, we showed that $\beta = 1/3$ in the limit. That means $\alpha = 1/6$. In other words, we find $I_{cm} = \frac{1}{6}m\ell^2$

REMARK: Notice that the moment of inertia of the shell comes out to be $I = m\ell^2/3$, that of a rod hinged at one end. We can see this most easily by rotating each of the outer rods by $\pi/2$, resulting in a rod with length ℓ and one end touching the axis of rotation. We can do this because we made no assumption of the orientation of the shape in the Parallel Axis Theorem.

Squashing provides an alternative, faster method for finding the moment of inertia of a square. Using the perpendicular axis theorem and symmetry, we find (Fig. 2.10):

$$I_z = I_x + I_y = 2I_c$$

Where I_c is the moment of inertia about any line that goes through the centroid. We will choose a line that is parallel to both sides of the square so that we can squash the square into a rod. We then see $I_c = \frac{1}{12}m\ell^2$. Solving the moment of inertia about the z -axis, we find $I_z = \frac{1}{6}m\ell^2$.

2.6 Rectangle

Solution 2: A rectangle is very similar to a square. All that changes is our use of the Parallel Axis Theorem.

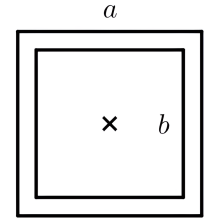
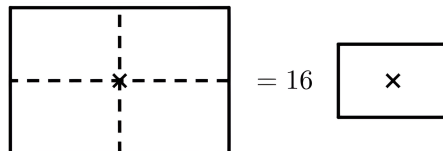


Figure 2.9

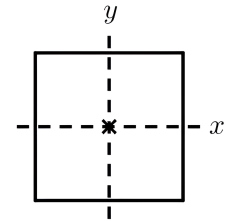
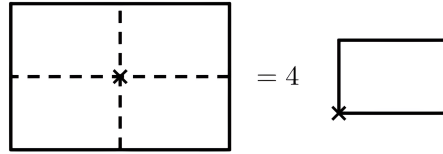


Figure 2.10



$$\boxed{\times} = \boxed{\times} + \frac{1}{4}m(a^2 + b^2)$$

Solving, we find $I_{cm} = \frac{m}{12}(a^2 + b^2)$

Solution 2: We claim that the moment of inertia of a rectangle only depends on the radius r of the circle that circumscribes the rectangle. By the Pythagorean Theorem:

$$r^2 = (a/2)^2 + (b/2)^2 = (a^2 + b^2)/4$$

To show this, we cut the rectangle into N rectangular slices. We then seek to find the mass of the outermost slice. We can begin by finding the surface density (Fig. 2.11) .

$$\sigma = \frac{m}{A} = \frac{m}{ab}$$

In the outermost slice, the combined left and right rods have a width b/N and a height a .

$$m_1 = \sigma(ab/N) = m/N$$

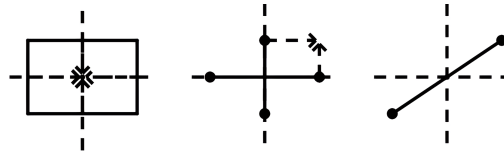
The combined top and bottom rods have a width b and a height a/N .

$$m_2 = \sigma(ab/N) = m/N$$

We see that the top and bottom rods have the same mass as the left and right rods despite their differing lengths $m_1 = m_2 \equiv m$.

We can then apply the Parallel Axis Theorem to express the moment of inertia of the outermost slice in terms of rods located at the origin and two points located at the center of mass of each rod.

In a similar spirit as the Perpendicular Axis Theorem, we can move any two points to the diagonal of the rectangle (cutting the mass in half). In essence, we can diagonalize the outermost slice to get a rod with a point at the very end.



We see that the moment of inertia only depends on the length of the diagonal as we can diagonalize any slice. The resulting slice is a rod with length $2r$ and a point located a distance r away from the origin. A quick calculation tells us the coefficient of the outermost slice.

$$I_{\text{slice}} = \frac{1}{12}(m/2)(2r)^2 + (m/2)(r)^2 = \frac{2}{3}mr^2$$

We can then finally apply the coefficient relationship derived in the previous problem $\beta = 2\alpha$. We find $\alpha = 1/3$ meaning the moment of inertia becomes $I_{cm} = \frac{1}{3}mr^2$.

REMARK: We first showed that the mass of the top-bottom rods was the same as the mass of the left-right rods. We can see this easily using a square. If we cut the square into shells, then

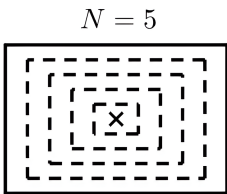


Figure 2.11

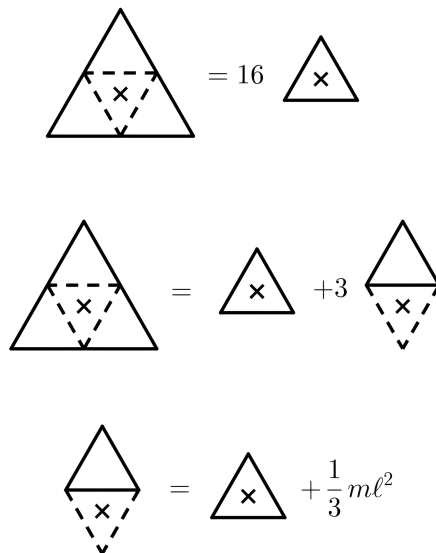
obviously the top, bottom, left, and right sides of the shell have the same mass. A rectangle is formed when we stretch a square, meaning the top and bottom rods increase in length (become longer) while the left and right rods increase in width (become thicker). Naturally, stretching doesn't change mass, only the area in which the mass lies.

We were able to diagonalize the two rods into one uniform length rod because the top-bottom and left-right rods had the same mass. But we had to cut the mass in half, otherwise we would be double-counting. An easy way to see this is to consider two mass m and m located at $x = r$ and $y = r$. Relative to the z -axis, they contribute a total moment of inertia $2mr^2$. Now consider a single mass m located at $(x, y) = (r, r)$. Relative to the z -axis, it contributes a moment of inertia $2mr^2$. It has half the total mass but contributes the same moment of inertia. Hence, when we diagonalized the rods, in order to maintain the moment of inertia, we had to cut the mass in half.

As a final statement, we could've solved for the moment of inertia using the perpendicular axis theorem and squashing. We see that the $1/12$ coefficient for a rectangle correctly comes from that of a rod.

2.7 Equilateral Triangle

Solution 1: We create our system of equations using scaling arguments, superposition, and the parallel axis theorem.



Solving, we find $I_{cm} = \frac{1}{12} m \ell^2$

Solution 2: As we've already established $\beta = 2\alpha$ for 2D shapes in moment of inertia, we simply need to find β , the coefficient of moment of inertia for the limiting shapes, a triangular shell.

A triangular shell consists of three rods (or one dense rod) at a distance $\ell/2\sqrt{3}$ away from the center of mass. We can apply the Parallel Axis Theorem.

$$I = md^2 + I_{cm} = m(\ell/2\sqrt{3})^2 + \frac{1}{12} m \ell^2 = \frac{1}{6} m \ell^2$$

Hence, $\beta = 1/6$ in the limit. That means $\alpha = 1/12$ resulting in a center of mass moment of inertia $I_{cm} = \frac{1}{12} m \ell^2$

REMARK: If you are confused about $\alpha = 2\beta$ or $\ell/2\sqrt{3}$, revisit Square and Equilateral Triangle (from the center of mass chapter).

2.8 Isosceles Triangle

Solution 1: As always, we construct our system of equations (this time with one more equation).

$$= 16 \times$$

$$= \times + \times + 2 \times$$

The next two require finding the distances between the center of masses. We can make this easier by drawing a lattice grid on top of our scaled triangle (Fig. 2.12).

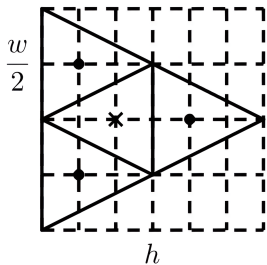


Figure 2.12

$$= \times + \frac{4}{9}mh^2$$

$$= \times + \frac{1}{9}mh^2 + \frac{1}{4}mw^2$$

Solving the system gives us $I_{cm} = \frac{1}{24}mw^2 + \frac{1}{18}mh^2$

Solution 2: We can begin with an equilateral shell. In the limit as the number of shells goes to infinity, the outermost shell becomes three rods with equal mass. Stretching the equilateral triangle to become an isosceles triangle doesn't change the mass. Hence, the three rods that form the outermost shell of an isosceles triangle have the same mass.

Now we have to find the coefficient of the moment of inertia of the outermost shell. Each rod has a mass $m/3$ and we'll have to apply the parallel axis theorem for each rod. Let the origin $(0,0)$ lie at the center of the shortest side. Then, the center of mass of the longer rods lies at $(h/2, \pm w/4)$ while the center of mass lies at $(h/3, 0)$ (Fig. 2.13).

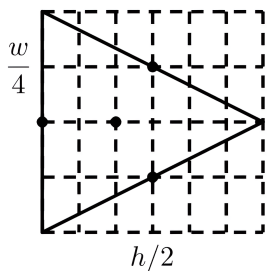


Figure 2.13

We can now find the moment of inertia using repeated use of the Parallel Axis Theorem. Note that the short rod m_1 has a length w while the longer rods m_2 has a length $\sqrt{h^2 + (w/2)^2}$ by the Pythagorean Theorem.

$$I_1 = \left(\frac{m}{3}\right) \left[\left(\frac{h}{3}\right)^2 + \frac{1}{12}w^2 \right]$$

$$I_2 = \left(\frac{2m}{3}\right) \left[\left(\frac{h}{2} - \frac{h}{3}\right)^2 + \left(\frac{w}{4}\right)^2 + \frac{1}{12} \left[h^2 + \left(\frac{w}{2}\right)^2 \right] \right]$$

We find the moment of inertia of the shell to be $I_{shell} = I_1 + I_2 = \frac{1}{12}mw^2 + \frac{1}{9}mh^2$. We have two betas $\beta_1 = 1/12$ and $\beta_2 = 1/9$. We know the limiting relationship $\beta = 2\alpha$

so we can solve for both $\alpha_1 = 1/24$ and $\alpha_2 = 1/18$. Putting it together, we find

$$I_{cm} = \frac{1}{24}mw^2 + \frac{1}{18}mh^2$$

REMARK: Although our work may seem dubious, the Perpendicular Axis Theorem offers a compelling rational. We found the moment of inertia of the shell relative to the z -axis. Relative to the x -axis, we could squash the triangle meaning h would play no role. Hence, $I_x = mw^2/12$. Similarly, relative to the y -axis, we could squash the triangle meaning w would play no role. Hence, $I_y = mh^2/18$. Relative to each axis, the limiting relationship we derived in square still applies. That means, we can find the moment of inertia of the solid triangle about each axis, and recombine them using the Perpendicular Axis Theorem.

2.9 Circle

Solution 1: A circle is merely the superposition of many thin isosceles triangles that are concurrent at the apex. We find the moment of inertia of an isosceles triangle about the apex using the previous problems result and the Parallel Axis Theorem.

$$I_{apex} = m(2h/3)^2 + mh^2/18 + mw^2/24$$

We find $I_{apex} = mh^2/2 + mw^2/24$. In the limit of infinitely many isosceles triangle, the width goes to zero $w \ll h$ while the height approaches the radius of the circle $h \rightarrow r$. The moment of inertia then becomes $I = \frac{1}{2}mr^2$

Solution 2: We can take the difference of two circles to get a hoop-like shape. In the limit as the two circles become each other, their difference does become a hoop with moment of inertia $I = \beta mr^2 = mr^2$. That means, $\beta = 1$. In Square, we derived the coefficient relationship between two 2D shapes $\beta = 2\alpha$. Hence, $\alpha = 1/2$ and the moment of inertia of a solid circle is $I = \frac{1}{2}mr^2$.

2.10 Cylinder

Consider a cylinder with mass m and radius r . We can cut it into horizontal slices (Fig. 2.14) Δm_h with equal width and relate the total moment of inertia $mr^2/2$ to the sum of the slices as

$$\frac{1}{2}mr^2 = \sum \Delta I_h$$

Now, we revisit our original cylinder and cut it into vertical slices Δm_v (Fig. 2.15). We find the total moment of inertia to be

$$I = \sum \Delta I$$

We can then apply the parallel axis theorem to each slice to find

$$I = \sum (\Delta mx^2 + \Delta I_v)$$

where ΔI_v is the moment of inertia about a vertical axis through the slice.

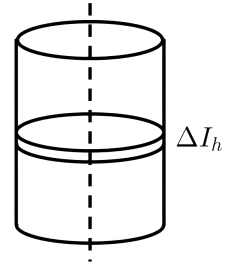
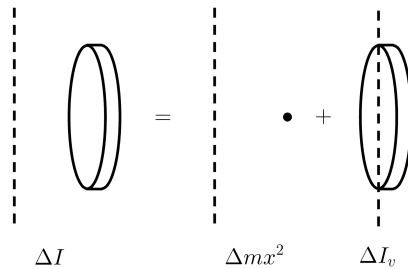


Figure 2.14

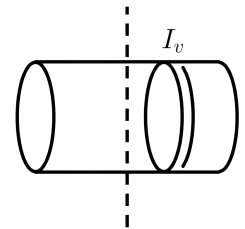


Figure 2.15

Then, recall using the Perpendicular Axis Theorem alongside symmetry, we can show

$$\Delta I_h = 2\Delta I_v$$

Additionally, notice that Δmx^2 is the moment of inertia of the slice of a rod because each Δm has the same mass. The rod formed by the centers of each vertical slice has a length h . We then can substitute:

$$I = \sum \Delta mx^2 + \sum \frac{1}{2} \Delta I_h$$

We see that the total moment of inertia is $I = \frac{1}{12}mh^2 + \frac{1}{4}mr^2$

REMARK: The majority of the reasoning follows from the questions and exercises. Review them if you are confused.

2.11 Solid Cone

Let a larger solid cone S_1 have a major length a and a smaller solid cone S_2 have a major length b . Then let their difference, a conical frustum, be S_3 . We find their masses to be

$$m_1 = m \quad m_2 = \frac{b^3}{a^3}m \quad m_3 = \left(1 - \frac{b^3}{a^3}\right)m$$

In the limit as $S_2 \rightarrow S_1$, S_3 becomes a flat circle located a distance h away from the axis of rotation (Fig. 2.16). Quick use of the parallel axis theorem shows

$$I_3 = md^2 + I_{cm} = mh^2 + \frac{1}{4}mr^2$$

We then let the moment of inertia of a solid cone take the form $I = \alpha mr^2$ and the difference of two cones take the form $I = \beta mr^2$. Finally, we apply the superposition relationship $I_1 = I_2 + I_3$ as usual and take the limit.

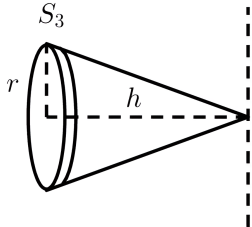


Figure 2.16

$$\begin{aligned} \lim_{b \rightarrow a} \beta \left(1 - \frac{b^3}{a^3}\right) mb^2 &= \lim_{b \rightarrow a} \left[\alpha ma^2 - \alpha \left(\frac{b^3}{a^3}\right) mb^2 \right] \\ \lim_{b \rightarrow a} \beta b^2 \left(\frac{a^3 - b^3}{a^3}\right) &= \lim_{b \rightarrow a} \alpha \left(\frac{a^5 - b^5}{a^3}\right) \\ \lim_{b \rightarrow a} \beta b^2 (a - b)(a^2 + ab + b^2) &= \lim_{b \rightarrow a} \alpha (a - b)(a^4 + ab^3 + a^2b^2 + ab^3 + b^4) \\ \lim_{b \rightarrow a} \beta b^2 (a^2 + ab + b^2) &= \lim_{b \rightarrow a} \alpha (a^4 + ab^3 + a^2b^2 + ab^3 + b^4) \\ \beta a^2 (a^2 + a * a + a^2) &= \alpha (a^4 + a * a^3 + a^2 * a^2 + a * a^3 + a^4) \end{aligned}$$

Solving, we find $3\beta = 5\alpha$. Applying the result to the limit; $\beta_1 = 1$ and $\beta_2 = 1/4$, we find $\alpha_1 = 3/5$ and $\alpha_2 = 3/20$. That means the moment of inertia of a solid cone is $I = \frac{3}{5}mh^2 + \frac{3}{20}mr^2$

REMARK: The difference of two solid cones can be a hollow cone; however, we choose the two solid cones to be aligned at their apex so that their coefficient α would be the same. That means, their differences ends up as a conical frustum. Had you instead chose to align the cones at their bases with the moment of inertia calculated with respect to that axis, then the difference of the two cones would indeed be a hollow cone.

2.12 Spherical Shell

Solution 1: We begin by cutting the sphere into horizontal slices Δm_h (Fig. 2.17), establishing

$$I = \sum \Delta I_h$$

Then, we cut the sphere into vertical slices Δm_v and use the Parallel Axis Theorem, establishing

$$I = \sum \Delta I = \sum (\Delta mx^2 + \Delta I_v)$$

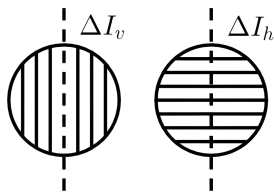


Figure 2.17

Where ΔI_v is the moment of inertia of a slice about a diametric axis. The Perpendicular Axis Theorem and symmetry tells us $\Delta I_h = 2\Delta I_v$. Then, recall that the mass of each slice in a hollow sphere is constant meaning Δm is constant.

That means, $\Delta m x^2$ is the moment of inertia of the slice of a uniform density rod. As the spherical shell has a radius r , the length of the rod would be $2r$. We then substitute.

$$I = \sum \Delta m x^2 + \sum \frac{1}{2} \Delta I_h$$

$$I = \frac{1}{12} (m) (2r)^2 + \frac{1}{2} I$$

Solving for I , we find $I = \frac{2}{3} m r^2$

REMARK: If you're confused why the mass of each slice is constant, revisit Hollow Hemisphere (from the Center of Mass Chapter).

Solution 2: We can apply the Three Axes Theorem with the origin at the center of the sphere. All the mass on the sphere is located a distance r away from the origin. We find

$$\int r^2 dm = m r^2$$

Additionally, the sphere is symmetric meaning the moment of inertia about any axis will be the same.

$$I_x = I_y = I_z \equiv I$$

We then substitute into the Three Axes Theorem.

$$I + I + I = 2 * m r^2$$

We find the moment of inertia to be $I = \frac{2}{3} m r^2$

2.13 Solid Sphere

Solution 1: We can utilize the limiting coefficient relationship derived in solid cone, $3\beta = 5\alpha$. A spherical shell is the difference of two solid spheres, meaning we can apply the relationship. The moment of inertia of a spherical shell is $I = 2mr^2/3$ meaning $\beta = 2/3$. We find $\alpha = 2/5$ resulting in the moment of inertia of a solid sphere being $I = \frac{2}{5} m r^2$

Solution 2: We seek to apply the Three Axes Theorem. We can cut the solid sphere into thin cones and superpose them all. Now, finding $\int r^2 dm$ is the same as finding the moment of inertia of a thin solid cone.

In Solid Cone, we had shown $I_{cone} = \frac{3}{5} m r^2 + \frac{3}{20} m h^2$. Taking the limit $h \ll r$ results in a thin cone or

$$I = \frac{3}{5} m r^2$$

We finally see

$$\int r^2 dm = \frac{3}{5} m r^2$$

By symmetry, the moment of inertia of a solid sphere is the same about any axis. We then know

$$I_x = I_y = I_z \equiv I$$

Substituting into the Three Axes Theorem, we find

$$I + I + I = 2 * \frac{3}{5} m r^2$$

Solving, we find $I = \frac{2}{5}mr^2$

REMARK: Recall r in the Three Axes Theorem represents the radial distance from the origin which we chose to be the center of the solid sphere. However, when we superposed all the thin cones into the equivalent of a line, the radial distance from the origin became the same as the perpendicular distance away from an axis perpendicular to the line. That's why finding the moment of inertia was a legal substitute for $\int r^2 dm$.